

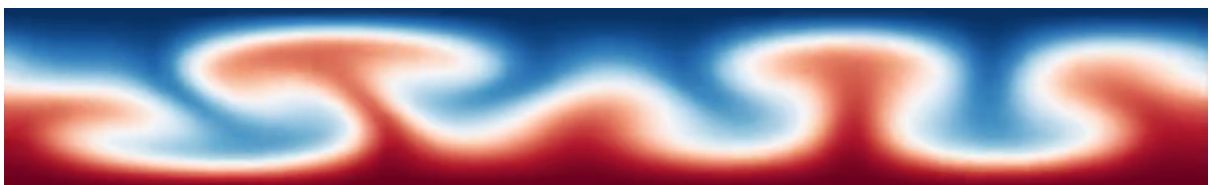


Adjoint-based Maximisation of Heat Flux in Models of Stellar Convection

Extended Version

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Abstract

This paper focuses on the application of the adjoint-based maximisation method to optimise systems of Rayleigh-Bénard convection. We measure the amount of convective heat transfer in the system, measured by the Nusselt number, Nu . We use the Dedalus v3 framework, a spectral PDE solver, to simulate these systems. We focus on optimising the domain width and initial conditions of our system, and the resulting astrophysical implications. We begin by proving how the method works, followed by some simple examples, and then apply it to our Rayleigh-Bénard convection system. We add complexities to our system, including the effects of rotation, to model astrophysical fluids at a large scale.

Introduction

Convection is an important natural process that governs the transport of energy through fluid dynamical systems, including stellar interiors, planetary atmospheres, and many engineering applications. Simulating these highly turbulent astrophysical environments requires sophisticated mathematical and computational modelling. This paper applies the adjoint-based maximisation method to optimise systems of Rayleigh-Bénard convection. By leveraging this method, we aim to maximise the amount of convective heat transfer within the system, parametrised by the Nusselt number, Nu , and find an appropriate scaling between this Nusselt number and the Rayleigh number.

To numerically simulate these complex fluid dynamics, this research utilises the Dedalus v3 framework, an open-source spectral PDE solver. The primary focus of this optimisation is the system's domain width and initial conditions, with the ultimate goal of modelling fluid behaviours that have broader astrophysical implications.

The paper is structured as follows: Chapter 1 establishes the mathematical foundation by proving the adjoint-based method for both linear and non-linear cases, accompanied by one-dimensional and two-dimensional examples that highlight the method's computational advantages over brute-force gradient calculations. Chapter 2 applies this framework directly to Rayleigh-Bénard problems. We define the governing Navier-Stokes equations under the Boussinesq approximation and investigate the scaling of the Nusselt number at various steady states. This section details the analytical setup for optimising domain width and initial conditions. Finally, Chapter 3 introduces rotational complexities vital for accurately modelling large-scale astrophysical fluids, specifically observing how the Coriolis force and Taylor number impact convective transport, focusing on the critical Rayleigh number.

Throughout the paper, we will explore physical implications, as well as the computational limitations of our methods. We aim to build a holistic view of this method and its application to our Rayleigh-Bénard problem.

Chapter 1

The Adjoint-Based Method

Before we can construct the problem this paper aims to solve, we need to prove the most general case of an Adjoint-Based Maximisation method and the algorithm required to solve it.

In this proof, we aim to extremise some objective function $J(\mathbf{y}, \mathbf{u})$ with:

$$\begin{aligned}\mathbf{y}(\mathbf{x}, t) &= (y_1(\mathbf{x}, t), y_2(\mathbf{x}, t), \dots, y_m(\mathbf{x}, t))^T \in \mathbb{R}^m \\ \mathbf{u}(\mathbf{x}, t) &= (u_1(\mathbf{x}, t), u_2(\mathbf{x}, t), \dots, u_p(\mathbf{x}, t))^T \in \mathbb{R}^p\end{aligned}$$

where $\mathbf{y}$ is our state vector, and $\mathbf{u}$ is our control vector. The state vector, $\mathbf{y}$, has $m - 1$ spatial dimensions, and we will choose the control vector, $\mathbf{u}$, to represent the initial conditions of our problem. We have picked this to make the problem setup more intuitive later. $\mathbf{u}(\mathbf{x}, t)$ does not have to be the initial conditions of the system; it can be involved directly in the evolution equation of the system, which we will see in a later example. It can even be a variable defining the size of our domain; however, this does require more work (as we will see).

Essentially, this method will find the control vector $\mathbf{u}$ that minimises (or maximises) some objective function, J , such that $\mathbf{y}$ obeys its evolution equation.

The underlying algorithm we will use to find the optimal $\mathbf{u}$ is a gradient-descent method defined by the following:

$$\mathbf{u}_{n+1} = \mathbf{u}_n - \alpha \nabla_{\mathbf{u}} J \tag{1.1}$$

where $\nabla_{\mathbf{u}} J$ is the total functional derivative (or Fréchet derivative) of J with respect to $\mathbf{u}$, and $\alpha > 0$ is some learning rate. Note that the gradient-descent algorithm minimises J ; so if we want to maximise J , we simply minimise $-J$. This algorithm requires that J is a suitably smooth function within our parameter space. We will see later that this is not always the case, depending on our numerical solver.

1.1 Proof of the Adjoint-Based Method

1.1.1 Linear Case

Let our time, $t \in [0, T]$, $T > 0$, and space, $\mathbf{x} \in \Omega \subset \mathbb{R}^{m-1}$, be defined in their respective domains. Let us also define the following inner product:

$$\langle\langle \mathbf{f}, \mathbf{g} \rangle\rangle = \int_0^T \int_{\Omega} \mathbf{f} \cdot \mathbf{g} d\Omega dt$$

For a general system involving first-order time derivatives, we define the evolution equation and initial conditions as follows:

$$\frac{\partial \mathbf{y}}{\partial t} + \mathcal{A}\mathbf{y} = \mathbf{f}(\mathbf{x}, t) \quad (1.2)$$

$$\mathbf{y}(\mathbf{x}, 0) = \mathcal{B}(u_1(\mathbf{x}), \dots, u_p(\mathbf{x}))^T \quad (1.3)$$

where $\mathcal{A}$ is a $m \times m$ linear operator acting on $\mathbf{y}$, $\mathcal{B}$ is an $m \times p$ linear operator, and $\mathbf{u}(\mathbf{x})$ is part of defining the initial condition that we want to optimise. It is important to consider some assumptions we now make. We assume that $\mathbf{y}$ is defined on $\Omega \times [0, T]$, and that we can find the respective adjoint operators of $\mathcal{A}$ and $\mathcal{B}$, meaning, for some vector fields $\mathbf{f}$ and $\mathbf{g}$, there exists a linear operator, $\mathcal{A}^*$, such that;

$$\langle\langle \mathbf{g}, \mathcal{A}\mathbf{h} \rangle\rangle = \langle\langle \mathcal{A}^*\mathbf{g}, \mathbf{h} \rangle\rangle \quad (1.4)$$

and similarly, $\mathcal{B}^*$ exists. It is crucial to note that the system of PDE's that we define in (1.2) is linear. If they are not, it requires an extra step to linearise the problem which we will cover when setting up our more complex problem in Chapter 2. For now, we will assume that we have a linear system of PDEs.

Now, let us construct the adjoint vector field, $\boldsymbol{\lambda}(\mathbf{x}, t) \in \mathbb{R}^m$ and some Lagrangian:

$$\mathcal{L}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) = J(\mathbf{y}, \mathbf{u}) + \int_0^T \int_{\Omega} \boldsymbol{\lambda} \cdot \left(\mathbf{f} - \frac{\partial \mathbf{y}}{\partial t} - \mathcal{A}\mathbf{y} \right) d\Omega dt = J(\mathbf{y}, \mathbf{u}) \quad (1.5)$$

Using integration by parts on the second and third term inside the double integral gives the results:

$$\begin{aligned} \int_0^T \int_{\Omega} \boldsymbol{\lambda} \cdot (-\mathcal{A}\mathbf{y}) d\Omega dt &= \langle\langle \boldsymbol{\lambda}, -\mathcal{A}\mathbf{y} \rangle\rangle, \text{ then using (1.4)} \\ &= \langle\langle -\mathcal{A}^*\boldsymbol{\lambda}, \mathbf{y} \rangle\rangle \\ &= \int_0^T \int_{\Omega} (-\mathcal{A}^*\boldsymbol{\lambda}) \cdot \mathbf{y} d\Omega dt \\ \int_0^T \int_{\Omega} \boldsymbol{\lambda} \cdot \left(-\frac{\partial \mathbf{y}}{\partial t} \right) d\Omega dt &= \int_0^T \int_{\Omega} \frac{\partial \boldsymbol{\lambda}}{\partial t} \cdot \mathbf{y} d\Omega dt - \int_{\Omega} (\boldsymbol{\lambda}(\mathbf{x}, T) \cdot \mathbf{y}(\mathbf{x}, T) - \boldsymbol{\lambda}(\mathbf{x}, 0) \cdot \mathbf{y}(\mathbf{x}, 0)) d\Omega \end{aligned}$$

Finding this adjoint operator, $\mathcal{A}^*$, will naturally lead to imposing boundary conditions $\boldsymbol{\lambda}$ must satisfy (see two-dimensional example). We then enforce the initial conditions (1.3) and substitute these results into (1.5), giving the following:

$$\begin{aligned} \mathcal{L} &= J(\mathbf{y}, \mathbf{u}) + \int_0^T \int_{\Omega} \left(\frac{\partial \boldsymbol{\lambda}}{\partial t} - \mathcal{A}^*\boldsymbol{\lambda} \right) \cdot \mathbf{y} d\Omega dt + \int_0^T \int_{\Omega} \boldsymbol{\lambda} \cdot \mathbf{f} d\Omega dt \\ &\quad - \int_{\Omega} (\boldsymbol{\lambda}(\mathbf{x}, T) \cdot \mathbf{y}(\mathbf{x}, T) - \boldsymbol{\lambda}(\mathbf{x}, 0) \cdot \mathcal{B}\mathbf{u}(\mathbf{x})) d\Omega \end{aligned}$$

To find a stationary point of J with respect to our state vector, we set the variation $\delta_{\mathbf{y}}\mathcal{L} = 0$. That is, perturbing $\mathbf{y}$ by an infinitesimal amount does not change the value of $\mathcal{L} = J$. For each iteration, $\mathbf{u}_n$ is a fixed vector field - the initial conditions - meaning $\delta\mathbf{y}(\mathbf{x}, 0) = \mathbf{0}$. By taking the first-order variation of $\mathcal{L}$ with respect to $\mathbf{y}$, we get:

$$\delta_{\mathbf{y}}\mathcal{L} = \mathcal{L}(\mathbf{y} + \delta\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) - \mathcal{L}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) \quad (1.6)$$

$$0 = \delta\mathbf{y} \frac{\partial\mathcal{L}}{\partial\mathbf{y}} \equiv \delta y_i \frac{\partial\mathcal{L}}{\partial y_i} \quad (1.7)$$

We must now assume that our objective function, J , is differentiable across our domain, meaning that it can be written in the form $J(\mathbf{y}, \mathbf{u}) = \int_0^T \int_{\Omega} F(\mathbf{y}, \mathbf{u}) d\Omega dt$, which allows us to write the following result:

$$0 = \int_0^T \int_{\Omega} \left(\partial_{\mathbf{y}} F + \frac{\partial\boldsymbol{\lambda}}{\partial t} - \mathcal{A}^* \boldsymbol{\lambda} \right) \cdot \delta\mathbf{y} d\Omega dt - \int_{\Omega} \boldsymbol{\lambda}(\mathbf{x}, T) \cdot \delta\mathbf{y}(\mathbf{x}, T) d\Omega \quad (1.8)$$

We now impose the boundary condition, $\boldsymbol{\lambda}(\mathbf{x}, T) = \mathbf{0}$, and by the Fundamental Lemma of Calculus of Variations, the first integrand must vanish. Hence, in index notation, our vector field, $\boldsymbol{\lambda}$, satisfies the following:

$$-\frac{\partial\lambda_i}{\partial t} + \mathcal{A}_{ij}^* \lambda_j = \frac{\partial F}{\partial y_i}, \quad \boldsymbol{\lambda}(\mathbf{x}, T) = \mathbf{0}, \quad i = 1, 2, \dots, m \quad (1.9)$$

We now have a system defining $\boldsymbol{\lambda}(\mathbf{x}, t)$, that satisfies the conditions necessary to find an extremum of our objective function, J . We now need to show how it can actually be used to find the optimal solution. We know the total variation of $\mathcal{L}$ is the sum of its variation arising from $\mathbf{y}$, $\mathbf{u}$ and $\boldsymbol{\lambda}$, meaning $\delta\mathcal{L} = \delta_{\mathbf{u}}\mathcal{L} + \delta_{\mathbf{y}}\mathcal{L} + \delta_{\boldsymbol{\lambda}}\mathcal{L}$. We already imposed $\delta_{\mathbf{y}}\mathcal{L} = 0$, and $\delta_{\boldsymbol{\lambda}}\mathcal{L} = 0$ is obtained by calculating $\frac{\partial\mathcal{L}}{\partial\boldsymbol{\lambda}}$ from (1.5). Hence, at the extremum, any variation is explained by our control vector, $\mathbf{u}$. All this is saying is that J is completely determined by $\mathbf{u}$, which is exactly what we want.

Using these results, we have that $\delta\mathcal{L} = \delta J = \delta_{\mathbf{u}}\mathcal{L}$. Now, from $\mathcal{L}$ we have:

$$\begin{aligned} \delta_{\mathbf{u}}\mathcal{L} &= \delta_{\mathbf{u}}J + \delta_{\mathbf{u}}(\mathcal{L} - J) \\ \delta J &= \int_{\Omega} \frac{\partial F}{\partial \mathbf{u}} \cdot \delta\mathbf{u} d\Omega + \int_{\Omega} \mathcal{B}^* \boldsymbol{\lambda}(\mathbf{x}, 0) \cdot \delta\mathbf{u} d\Omega \\ &= \int_{\Omega} \left(\frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^* \boldsymbol{\lambda}(\mathbf{x}, 0) \right) \cdot \delta\mathbf{u} d\Omega \end{aligned}$$

Note that we only integrate over the spatial domain because $\mathbf{u}$ is strictly defined as an initial condition. By definition, $\delta J = \int_{\Omega} \nabla_{\mathbf{u}} J \cdot \delta\mathbf{u} d\Omega$, and thus we have the result:

$$\nabla_{\mathbf{u}} J = \frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^* \boldsymbol{\lambda}(\mathbf{x}, 0) \quad (1.10)$$

We can substitute this into (1.1) to achieve the following form of the gradient-descent algorithm, which will form the backbone for solving our optimisation problems in this paper.

$$\mathbf{u}_{n+1} = \mathbf{u}_n - \alpha \left(\frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^* \boldsymbol{\lambda}(\mathbf{x}, 0) \right) \quad (1.11)$$

1.1.2 Non-Linear Case

It is important to consider what we do in the case that $\mathcal{A}$ is non-linear, as we will see in Chapter 2. If $\mathcal{A}$ is non-linear, we have to linearise the forward problem involving $\mathbf{y}$ to be able to find the adjoint problem. This is because we expand the variation with respect to $\mathbf{y}$ to the first order in (1.7) so that we can factorise out each expression in (1.8). This allows us to define the adjoint problem. If we were to expand the variation to n^{th} -order, the number of equations our system must satisfy increases with $\mathcal{O}(m^n)$. This would allow us to use more advanced optimisation algorithms, like Newton's method, but at great computational cost. For the gradient-descent/ascent algorithm, we only require the first-order variation of J anyway, as we are only calculating a first derivative. Therefore, we only make the problem as complicated as it needs to be and expand the problem to first order, resulting in a linear adjoint problem. In practice, this is sufficiently accurate in many cases.

1.2 The Adjoint-Based Algorithm

At this point, we have proven that we can evaluate the analytic gradient $\frac{\partial F}{\partial \mathbf{u}} + \mathcal{B}^* \boldsymbol{\lambda}(\mathbf{x}, 0)$ instead of approximating $\nabla_{\mathbf{u}} J$ via finite differences to iterate through control vectors, $\mathbf{u}_n$. Calculating $\frac{\partial F}{\partial \mathbf{u}}$ is far easier than calculating a total derivative, as it is usually something we can do analytically. This is because we do not need to calculate $\frac{\partial J}{\partial \mathbf{x}} \frac{d\mathbf{x}}{d\mathbf{u}}$ across our domain, which would be computationally expensive.

To carry out the adjoint-based maximisation algorithm [7] for the general use-case of $\mathbf{u}_n$, we need to carry out the following steps:

1. Start with an initial guess, $\mathbf{u}_0(\mathbf{x})$, for the optimal initial conditions.
2. Forward solve the system (1.2) using $\mathbf{u}_n(\mathbf{x})$ to define an updated problem, then find $\mathbf{y}_n(\mathbf{x}, t)$.
3. Evaluate $J(\mathbf{y}_n, \mathbf{u}_n)$. If $|J(\mathbf{y}_n, \mathbf{u}_n) - J(\mathbf{y}_{n-1}, \mathbf{u}_{n-1})| < \epsilon$, some tolerance, then stop.
4. Otherwise, solve the system (1.9) going backwards in time (starting with $t = T$) to find the adjoint field $\boldsymbol{\lambda}(\mathbf{x}, t)$. In practice, we let $\tau = T - t$ and solve for $\tau \in [0, T]$.
5. We then use the adjoint field's 'final' state of $\boldsymbol{\lambda}(\mathbf{x}, t = 0)$ to update $\mathbf{u}_n$.
6. Repeat steps 2-5 until J converges, or for N iterations so $n = 1, 2, \dots, N$.

So far, this may seem rather abstract and theoretical, so I have worked through the previous proof for a simple one-dimensional example, where $\mathbf{u}$ does not represent the initial conditions, but is involved in the evolution equation of the system. This outlines the benefits of using this adjoint-based method over a brute-force method. I will then demonstrate a second example in higher dimensions, using $\mathbf{u}(\mathbf{x})$ to represent the system's initial conditions.

1.3 Examples

1.3.1 One-Dimension

This basic example does not include a time derivative term (the next example does), but still employs the adjoint-based optimisation method to minimise an objective function, $J(y, u)$, given some system defined by some linear operator and boundary conditions. Essentially, this solution finds the optimal function, $u(x)$, such that $y(x)$ satisfies the ODE and is as 'close' to $y_{target}(x)$ as possible. To avoid loss of generality, we define the system as follows.

$$Ly \equiv \frac{dy}{dx} + ay = u(x), \quad y(0) = 0, \quad x \in [0, L], \quad a \in \mathbb{R} \quad (1.12)$$

$$J = \frac{1}{2} \int_0^L (y(x) - y_{target}(x))^2 dx \quad (1.13)$$

Note that $\mathcal{A} = a$ in this example, which is self-adjoint (so $\mathcal{A}^* = a$). We want to minimise $J(y, u)$, so we will be using the gradient-descent algorithm to iterate the function u , N times. That is,

$$u_{n+1} = u_n - \alpha \nabla_u J$$

for $n = 0, 1, \dots, N$, where u_0 is some initial guess, and α is some constant learning rate. Firstly, it is important to note that by definition:

$$\delta_u J = \int_0^L \nabla_u J(x) \delta u(x) dx$$

Now construct the Lagrangian, $\mathcal{L} = J$, for which we want its variation to be zero, and some function $\lambda(x)$.

$$\mathcal{L} = \frac{1}{2} \int_0^L (y - y_{target})^2 dx + \int_0^L \lambda(x) \left(u(x) - \frac{dy}{dx} - ay \right) dx = J$$

Using integration by parts: $\int_0^L \lambda(x) \frac{dy}{dx} dx = [\lambda(x)y(x)]_0^L - \int_0^L y \frac{d\lambda}{dx} dx$

$$\implies \mathcal{L} = \int_0^L \left(\frac{1}{2} (y - y_{target})^2 + y \left(\frac{d\lambda}{dx} - a\lambda \right) + \lambda u \right) dx - \lambda(L)y(L) + \lambda(0)y(0)$$

Now, impose $\lambda(L) = 0$, and set the variation with respect to y , $\delta_y \mathcal{L} = 0$ for some arbitrary δy :

$$\begin{aligned} \delta_y \mathcal{L} &= \delta y \frac{d\mathcal{L}}{dy} + \mathcal{O}((\delta y)^2) = 0 \\ \implies \delta y \frac{d\mathcal{L}}{dy} &= 0 = \delta y \frac{d}{dy} \int_0^L \left(\frac{1}{2} (y - y_{target})^2 + y \left(\frac{d\lambda}{dx} - a\lambda \right) + \lambda u \right) dx \\ 0 &= \delta y \int_0^L \frac{\partial}{\partial y} \left(\frac{1}{2} (y - y_{target})^2 + y \left(\frac{d\lambda}{dx} - a\lambda \right) + \lambda u \right) dx \\ \implies 0 &= \int_0^L \left(y - y_{target} + \frac{d\lambda}{dx} - a\lambda \right) \delta y dx \end{aligned}$$

By the Fundamental Lemma of Calculus of Variations, the integrand must vanish:

$$\implies -\frac{d\lambda}{dx} + a\lambda = y - y_{target}, \quad \lambda(L) = 0 \quad (1.14)$$

Note that in this example, $F = \frac{1}{2}(y - y_{target})^2$, which now appears in (1.14) via $\frac{\partial F}{\partial y}$. This evolution equation for λ is in the form seen in (1.9). This also means that we would expect only a λ -term to appear in the $\nabla_u J$ term as $\frac{\partial J}{\partial u} = 0$. Now, let us focus on the variation in the u -component to first order δu :

$$\begin{aligned} \delta_u \mathcal{L} &= \delta u \frac{d\mathcal{L}}{du} + \mathcal{O}((\delta u)^2) = \delta_u J \\ &= \delta u \frac{d}{du} \int_0^L \left(\frac{1}{2} (y - y_{target})^2 + y \left(\frac{d\lambda}{dx} - a\lambda \right) + \lambda u \right) dx \\ &= \delta u \int_0^L \frac{\partial}{\partial u} \left(\frac{1}{2} (y - y_{target})^2 + y \left(\frac{d\lambda}{dx} - a\lambda \right) + \lambda u \right) dx \\ \delta_u J &= \int_0^L \lambda(x) \delta u(x) dx = \int_0^L \nabla_u J \delta u(x) dx \\ &\implies \nabla_u J = \lambda(x) \end{aligned}$$

This is in the form we expected. Hence, we can use $\lambda(x)$ instead of $\nabla_u J$ when iterating u_n , which is advantageous because solving for $\nabla_u J$ via brute-force would be computationally impractical. When numerically solving this problem, we achieve the following results:

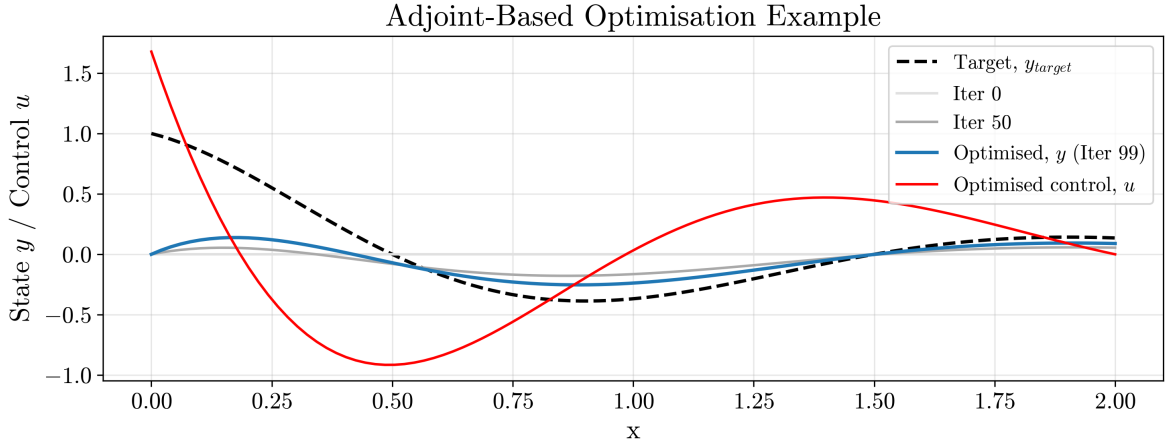


Figure 1.1: Plot of the optimised control vector, $\mathbf{u}$, and state vector, $\mathbf{y}_n$, against iteration number.

1.3.2 Advantages over a Brute-Force Method

Using the adjoint-based method on a discretised domain to extremise an objective function is computationally more efficient compared to a brute-force method. For each iteration of u_n with the adjoint-based method, we require one forward pass and one backward pass; we only have to solve two PDE systems over our time domain. Whereas using a brute-force method requires that we complete a forward pass for every (discrete) point on the control function to calculate the gradient approximation $\frac{dJ}{du}$ on each iteration of

u_n . In large spatial domains, a brute-force method is computationally impractical due to processing and storage constraints.

Our discretised domain is formed of k x -values, x_i , $i = 1, 2, \dots, k$. On each iteration of u_n , the brute-force method requires:

1. For the point on the grid $u_n(x_i) = u_i$, calculate an initial value $J_{n,i}$.
2. Perturb the point on the grid by a small amount, ϵ .
3. Calculate a new value $J_{n+1,i}$.
4. Calculate $\frac{\partial J}{\partial u_i} \approx \frac{J_{n+1,i} - J_{n,i}}{\epsilon}$.
5. Return u_i to where it was, and repeat 1-4 for every u_i .
6. Transform each point $u_i \rightarrow u_i - \alpha \frac{\partial J}{\partial u_i} \approx u_i - \alpha \frac{J_{n+1,i} - J_{n,i}}{\epsilon}$ to give the points on u_{n+1} .

Thus, each iteration of u_n would otherwise require solving for y (a forward pass), k times. Whereas, if we use the adjoint-based method, we do one forward pass and then one backward pass (solve for λ where λ satisfies (1.14)) per iteration of u_n . That is, the brute-force method would require kN passes, whereas using the adjoint-based method simply requires $2N$ passes over N iterations.

1.3.3 Two-Dimensions

In this example, I will focus less on the rigorous proof behind the theory and more on the computational practice of finding adjoint operators and deriving appropriate boundary conditions for the adjoint field, λ . We will define a two-dimensional (one spatial and one temporal) problem describing heat diffusivity.

Let us have a wall between two regions of fixed temperature, and we want to maximise the heat flux through the wall. We define our state vector (scalar temperature in this case), $\mathbf{y} = y(x, t)$, and control vector, $\mathbf{u} = u(x)$, where $x \in [0, L]$ and $t \in [0, T]$. Of course, using three spatial dimensions would lead to a more comprehensive understanding of this problem, but for simplicity, we will use one spatial dimension for now. In this example, we define the system as follows:

$$\begin{aligned} \frac{\partial y}{\partial t} - \nu \frac{\partial^2 y}{\partial x^2} &= 0 \\ y(x, 0) &= u(x) \\ y(L, t) &= 0, \quad y(0, t) = 100 \end{aligned}$$

with our objective function including the heat flux, and some weighted term to stop the optimiser setting $u(x, t) = \pm\infty$.

$$J = \int_0^T \int_0^L \left(-\nu \frac{\partial y}{\partial x} + \frac{\gamma}{2} (u(x))^2 \right) dx dt$$

Notice this is in the form of $J(\mathbf{y}, \mathbf{u}) = \int_0^T \int_\Omega F(\mathbf{y}, \mathbf{u}) d\Omega dt$. We can also notice that the linear operator, $\mathcal{A} = -\frac{\partial^2}{\partial x^2}$, and that $\mathcal{B} = 1$ in this case. Both are trivially self-adjoint, but we will need to formally show that $\mathcal{A}$ is self-adjoint to work out the appropriate boundary conditions needed to solve for $\lambda(x, t)$.

As we have done in the proof and previous example, let

$$\mathcal{L} = J(y, u) + \int_0^T \int_0^L \lambda \left(-\frac{\partial y}{\partial t} + \frac{\partial^2 y}{\partial x^2} \right) dx dt \quad (1.15)$$

Now using integration by parts:

$$\begin{aligned} \int_0^T \int_0^L \lambda \frac{\partial^2 y}{\partial x^2} dx dt &= \int_0^T \left[\lambda \frac{\partial y}{\partial x} \right]_0^T dt - \int_0^T \int_0^L \frac{\partial \lambda}{\partial x} \frac{\partial y}{\partial x} dx dt \\ &= \int_0^T \left[\lambda \frac{\partial y}{\partial x} - \frac{\partial \lambda}{\partial x} y \right]_0^L dt + \int_0^T \int_0^L y \frac{\partial^2 \lambda}{\partial x^2} dx dt \end{aligned}$$

This verifies that $\mathcal{A}$ is, in fact, self-adjoint subject to λ fulfilling some boundary conditions. We need

$$\left[\lambda \frac{\partial y}{\partial x} - \frac{\partial \lambda}{\partial x} y \right]_0^L = 0$$

We do not have homogeneous Dirichlet or Neumann boundary conditions, meaning we must use Calculus of Variations to derive the boundary conditions for λ . We mentioned in the proof that at the boundary conditions of our state vector, its variation is zero. That is

$$y(0, t) = 100 \implies \delta y(0, t) = 0 \quad \text{and} \quad y(L, t) = 0 \implies \delta y(L, t) = 0$$

The equations defining this system must also hold if we were to perturb y some small amount, namely, they must hold for $y^* = y + \delta y$. As derivatives are associative, we know that δy must also satisfy our system of equations. This means that we can have

$$\begin{aligned} \left[\lambda \frac{\partial \delta y}{\partial x} - \frac{\partial \lambda}{\partial x} \delta y \right]_0^L &= 0 \\ \lambda(L, t) \frac{\partial(\delta y)}{\partial x}(L, t) - \frac{\partial \lambda}{\partial x}(L, t) \delta y(L, t) - \lambda(0, t) \frac{\partial(\delta y)}{\partial x}(0, t) + \frac{\partial \lambda}{\partial x}(0, t) \delta y(0, t) &= 0 \\ \lambda(L, t) \frac{\partial(\delta y)}{\partial x}(L, t) - \lambda(0, t) \frac{\partial(\delta y)}{\partial x}(0, t) &= 0 \end{aligned}$$

Now, we can impose the boundary conditions $\lambda(L, t) = \lambda(0, t) = 0$ necessary for $\mathcal{A}$ to have an adjoint operator. From the other term in the integral in (1.15), we have that

$$\begin{aligned} \int_0^T \int_0^L -\lambda \frac{\partial y}{\partial t} dx dt &= \int_0^T \int_0^L \frac{\partial \lambda}{\partial t} y dx dt - \int_0^L (\lambda(x, T) y(x, T) - \lambda(x, 0) y(x, 0)) dx \\ \implies \mathcal{L} &= J(y, u) + \int_0^T \int_0^L \left(\frac{\partial \lambda}{\partial t} - \frac{\partial^2 \lambda}{\partial x^2} \right) y dx dt \\ &\quad - \int_0^L (\lambda(x, T) y(x, T) - \lambda(x, 0) y(x, 0)) dx \end{aligned}$$

We impose the 'initial' condition $\lambda(x, T) = 0$ and we know

$$\begin{aligned} F &= -\nu \frac{\partial y}{\partial x} + \frac{\gamma}{2} (u(x))^2 \\ \implies \frac{\partial F}{\partial y} &= -\nu \frac{\partial}{\partial y} \left(\frac{\partial y}{\partial x} \right) = -\nu \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial y} \right) = -\nu \frac{\partial}{\partial x} (1) = 0 \end{aligned}$$

Hence, using (1.9), we find that λ satisfies the system:

$$\begin{aligned} -\frac{\partial \lambda}{\partial t} + \frac{\partial^2 \lambda}{\partial x^2} &= 0 \\ \lambda(x, T) &= 0 \\ \lambda(L, t) = \lambda(0, t) &= 0 \end{aligned}$$

This system can be numerically (and analytically) solved. Solving for $\nabla_u J$ now becomes a simple task as a constant linear operator, $\mathcal{B}$, is trivially self-adjoint. We can now return to the original problem of minimising our objective function, J . From (1.11), we can iterate u_n via the iterative formula $u_{n+1}(x) = u_n(x) - \alpha(\gamma u_n(x) + \lambda(x, 0))$

$$\implies u_{n+1}(x) = (1 - \alpha\gamma)u_n(x) - \alpha\lambda(x, 0)$$

It may seem counter-intuitive to say the 'initial' condition of the adjoint field is at $t = T$, but this is because we are solving for λ going backwards in time - starting at $t = T$, and solving for the function $\lambda(x, 0)$.

We have now shown how we can optimise the initial conditions of a two-dimensional problem to maximise some arbitrary objective function. When solving this problem numerically, setting $\alpha = 0.05$, $\gamma = 1$, $\nu = 0.1$, and $u_0(x) = 100 - 50x$ we achieve the following results.

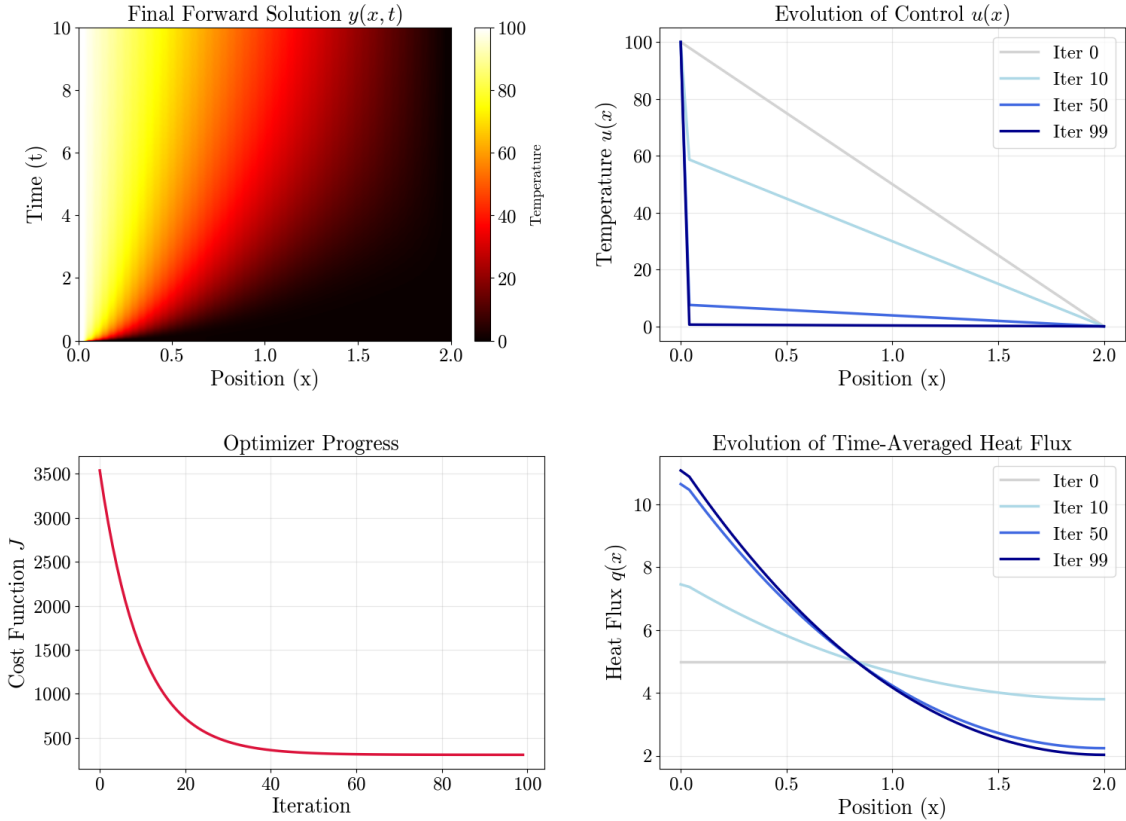


Figure 1.2: Two-Dimensional Adjoint Optimisation Results

Figure 1.2 shows that the optimal initial conditions to maximise the total heat flux = (integral under the time-averaged heat flux graph) $\times T$, across our domain tend towards

$u(x) = 0, x \in (0, L]$ and $u(0) = 100$. To verify this is the case, when we run the optimisation setting this as the initial conditions, J becomes constant. This makes sense mathematically as it creates a step at $x = 0$, giving an effectively 'infinite' flux at this point. Of course, setting $u_0(x) = 100, x \in [0, L)$ and $u_0(L) = 0$, would also create a step at $x = L$. However, the weighted $(u(x))^2$ term ensures that u is made as small as possible. In fact, the final initial condition this optimisation created, $u_{100}(x)$, is the best solution possible given the step size in x , δx . Choosing a smaller δx would result in initial conditions closer to the true 'best' initial conditions.

Chapter 2

Rayleigh-Benard Problems

We have proven the adjoint-based method is a viable method for optimising PDE problems, and gone through some examples; we can now begin to add complexity to construct the main problems this paper aims to solve.

We will be focusing on using the Navier-Stokes equations to model fluids within various geometries throughout these problems, focusing on the incompressible case. However, we will first cover the general compressible Navier-Stokes equations and then impose constraints suitable to our problems. In the most general sense, the Navier-Stokes equations (in 3-dimensions) are simply an application of Newton's Second Law, and other fundamental conservation laws. We define the fluid density $\rho(\mathbf{x}, t)$, fluid velocity $\mathbf{v}(\mathbf{x}, t)$, temperature $T(\mathbf{x}, t)$, pressure $p(\mathbf{x}, t)$, and $\mathbf{F}$ representing external forces, giving the following equations in Cartesian coordinates:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (\text{Conservation of Mass}) \quad (2.1)$$

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \nabla \cdot \tau + \mathbf{F} \quad (\text{Newton's Second Law}) \quad (2.2)$$

$$\rho c_v \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = -p(\nabla \cdot \mathbf{v}) + k \nabla^2 T + \text{tr}(\tau \nabla \mathbf{v}) \quad (\text{Advection-Diffusion Equation}) \quad (2.3)$$

where

$$\nabla \mathbf{v} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}, \quad \tau = \mu \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3} (\nabla \cdot \mathbf{v}) I \right)$$

These equations are well-known and well-studied [8], but very complicated to solve numerically, and often impossible to solve analytically. They define the constants for dynamic viscosity μ , thermal conductivity k , and specific heat capacity at a fixed volume, c_v .

Throughout these problems, we will also focus heavily on the Nusselt Number, Nu , defined as (heat transported by convection)/(heat transported by conduction). It is an important quantity in gaining an understanding of how heat is transported across a layer of fluid, and will be a quantity we aim to maximise in our problems.

In our first problem, we will apply a widely used approximation to the Navier-Stokes equations, and then build on this in the following problems.

2.1 Boussinesq Approximation

In our first problem, we will use the widely accepted Boussinesq approximation to the compressible Navier-Stokes equations [2]. It treats changes in the fluid's density as effectively negligible - essentially constant density. That is,

$$\frac{\partial \rho}{\partial t} = 0 \implies \nabla \cdot \mathbf{v} = 0 \quad \text{by (2.1)}$$

This then makes the rest of our equations much more manageable.

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{v} + \frac{1}{\rho} \mathbf{F} \quad (2.4)$$

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \kappa \nabla^2 T + \Phi \quad (2.5)$$

where Φ is some internal heating, and κ is the thermal diffusivity. In our problem, we will be setting $\Phi = 0$, as we focus on heating via the boundaries. We are now considering the incompressible Navier-Stokes equations, $\rho = \rho_0$.

To reduce the number of constants we're dealing with, we non-dimensionalise our equations by scaling each variable. We scale distance by the height of the layer of fluid, H , time by the thermal diffusion time, H^2/κ , and pressure by $\rho_0 H^2/\kappa$, as demonstrated in Goluskin, 2015 [2]. We then also use an approximation to the force created through buoyancy,

$$\frac{1}{\rho_0} \mathbf{F} = -g(1 - \beta(T - T_0)) \hat{\mathbf{e}}_z$$

where T_0 is the temperature of the top boundary, and β is the coefficient of thermal expansion. When we apply these to the above equations, we get the non-dimensionalised Boussinesq approximation equations:

$$\nabla \cdot \mathbf{v} = 0 \quad (2.6)$$

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\nabla p + Pr \nabla^2 \mathbf{v} + Pr Ra T \hat{\mathbf{e}}_z \quad (2.7)$$

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \nabla^2 T + Q \quad (2.8)$$

where Q is some internal heating that we will set to zero (but kept for now for the sake of generality). We also have defined some well-known constants, the Rayleigh and Prandtl numbers:

$$Ra := \frac{g \beta H^3 \Delta T}{\kappa \nu}$$

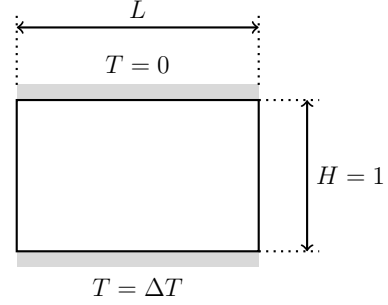
$$Pr := \frac{\nu}{\kappa}$$

where ν is kinematic viscosity, and ΔT is the difference in temperature between the two layers. The Rayleigh number is a measure of how much we are thermally driving the system - a large Rayleigh number indicates that the fluid is being largely influenced by buoyancy forces, $\mathbf{F}$. There is a minimum (critical) Rayleigh number, Ra_c , required for convection to occur for different boundary conditions. [10][6][1][9]

No-slip boundaries	$\mathbf{v}(x, y, 0, t) = \mathbf{v}(x, y, H, t) = \mathbf{0}$	$Ra_c \approx 1707.96$
Free-slip boundaries	$v_z(x, y, 0, t) = v_z(x, y, H, t) = 0$	$Ra_c \approx 657.511$
Mixed (one free-slip, one no-slip)	$\mathbf{v}(x, y, 0, t) = \mathbf{0}, v_z(x, y, H, t) = 0$	$Ra_c \approx 1100.65$

The Prandtl number is a material property of each fluid, and it is a measure of how quickly a fluid diffuses momentum compared to how fast it diffuses heat. A high Prandtl number represents a fluid that largely damps motion and is a poor conductor of heat, such as in the Earth's mantle. A small Prandtl number represents a fluid that weakly damps motion and conducts heat well, such as stellar plasmas.

Now we have built some intuition of the model defined in equations (2.6)-(2.8), we can begin to define the geometry for this problem. We will be focusing on a two-dimensional (x, z) , problem modelling a layer of fluid in a domain of height $H = 1$ and width L . The fluid will be heated from the bottom, and cooled at the top.



2.2 Steady States

When considering the steady state of a system, our equations (2.6)-(2.8) become far simpler as we set $\frac{\partial T}{\partial t} = 0$, $\frac{\partial \mathbf{v}}{\partial t} = \mathbf{0}$, meaning our non-dimensionalised Boussinesq equations become

$$\nabla \cdot \mathbf{v} = 0 \quad (2.9)$$

$$\mathbf{v} \cdot \nabla \mathbf{v} = -\nabla p + Pr \nabla^2 \mathbf{v} + Pr Ra T \hat{\mathbf{e}}_z \quad (2.10)$$

$$\mathbf{v} \cdot \nabla T = \nabla^2 T \quad (2.11)$$

which are far easier to solve, and completely remove our time dependency in this problem.

Investigating this behaviour numerically requires careful consideration of the spatial domain. Dedalus constructs the solution using a Fourier series with discrete wave numbers, $k_n = \frac{2\pi n}{L}$ for $n = 1, 2, 3, \dots, \hat{N}$. For a specific spatial mode k_n to actively contribute to the convection in Dedalus, the system's Rayleigh number must exceed that mode's specific marginal stability threshold ($Ra > Ra_c(k_n)$ - Equation (2.17)). However, at smaller values of L , we cannot fit in modes of a certain wave number. Dedalus is restricted to discrete wave numbers, so the critical Rayleigh number forms a step-function at narrow domain widths, as the fluid is forced to jump between non-optimal harmonic states. Similar physics exists for no-slip boundary conditions, though the boundaries prevent us from writing the solution down purely analytically.

2.2.1 Small Rayleigh numbers with free-slip boundaries

For small Rayleigh numbers, where the system is less turbulent, we can find an analytical approximation to our system at small Rayleigh numbers, which will allow us to find the critical Rayleigh number under free-slip boundary conditions. We will use the Galerkin truncation which assumes that we can express the fluid's temperature using three expressions: a linear background term, one rising/falling plume term ($\sin(\pi z)$), and one vertical distortion term ($\sin(2\pi z)$). At higher Rayleigh numbers, we require more

terms to express the temperature field, which is what Spectral Solvers like Dedalus accomplish. Eventually, this reaches a practical limit, preventing accurate simulations at higher Rayleigh numbers.

For Ra close to the critical Rayleigh number, Ra_c , we assume the following ansatz:

$$\psi(x, z) = A \sin(kx) \sin(\pi z), \quad T(x, z) = 1 - z + B \cos(kx) \sin(\pi z) + C \sin(2\pi z)$$

where k is the respective wave number, and $v_x = -\frac{\partial \psi}{\partial z}$, $v_z = \frac{\partial \psi}{\partial x}$. Notice that $1 - z$ is the conductive state of the system and is the trivial steady state of the system. Also, ψ immediately satisfies (2.9). We take the curl of the momentum equation, transforming it into a vorticity equation, which naturally gets rid of the pressure field, p , as we can eliminate it using the identity $\nabla \times \nabla p = \mathbf{0}$. Hence, under free-slip boundary conditions, (2.7) reduces to

$$0 = Pr \nabla^4 \psi + Pr Ra \frac{\partial T}{\partial x} \implies 0 = Pr(k^2 + \pi^2)^2 A - Pr Ra k B \quad (2.12)$$

Similarly, substituting our ansatz into (2.11), we find

$$-Ak \cos(kx) \sin(\pi z) + \frac{1}{2} AB k \pi \sin(2\pi z) - AC k \pi \cos(kx) \sin(\pi z) = \quad (2.13)$$

$$-(k^2 + \pi^2) B \cos(kx) \sin(\pi z) - 4\pi^2 C \sin(2\pi z) \quad (2.14)$$

and assuming equality of coefficients of corresponding terms on each side, we get

$$0 = kA - (k^2 + \pi^2)B + Ak\pi C \quad (2.15)$$

$$0 = \frac{k\pi}{2} AB + 4\pi^2 C \quad (2.16)$$

We have three equations in terms of three unknowns, which we can solve.

$$(2.12) \implies B = \frac{(k^2 + \pi^2)^2}{kRa} A$$

$$(2.16) \implies C = -\frac{k}{8\pi} AB \implies C = -\frac{(k^2 + \pi^2)^2}{8\pi Ra} A^2$$

$$(2.15) \implies 0 = kA - \frac{(k^2 + \pi^2)^3}{kRa} A - \frac{k(k^2 + \pi^2)^2}{8Ra} A^3$$

We can assume $A \neq 0$ as $A = 0$ corresponds to a stationary, purely conductive state. Hence, we get

$$1 - \frac{(k^2 + \pi^2)^3}{k^2 Ra} = \frac{(k^2 + \pi^2)^2}{8Ra} A^2 > 0$$

Notice how we now have a lower bound of the Rayleigh number, being our critical Rayleigh number, Ra_c . That is, for a convective ($A \neq 0$) state to exist,

$$Ra > \frac{(k^2 + \pi^2)^3}{k^2} = Ra_c \quad (2.17)$$

By differentiating with respect to k to minimise Ra_c , and setting this equal to zero at the critical wave number, k_c , we find

$$\begin{aligned} \frac{d}{dk} \left(\frac{(k^2 + \pi^2)^3}{k^2} \right) \Big|_{k=k_c} &= \frac{(4k_c^3 - 2k_c\pi^2)(k_c^2 + \pi^2)^2}{k_c^4} = 0 \\ \implies 4k_c^3 - 2k_c\pi^2 &= 0 \implies k_c^2 = \frac{\pi^2}{2} \implies Ra_c = \frac{27\pi^4}{4} \approx 657.511 \end{aligned}$$

This is a well-documented result [2], but we are interested in its impact on the Nusselt number. We define the Nusselt number as the ratio of heat transported via convection to that transferred by conduction; it is calculated as follows:

$$Nu = 1 + \langle v_z T \rangle$$

where v_z is the vertical component of a particles velocity, and $\langle \cdot \rangle$ represents an average across the domain [2]. When considering the ansatz of our field, we get

$$\langle v_z T \rangle = \frac{1}{L} \int_0^1 \int_0^L ABk \cos^2(kx) \sin^2(\pi z) dx dz = \frac{1}{4} ABk$$

This leads to the result that, for $Ra \sim Ra_c$ and $Ra \geq Ra_c$, the Nusselt number scales

$$Nu \approx 1 + 2 \left(1 - \frac{Ra_c}{Ra} \right) \quad (2.18)$$

As the system becomes more turbulent, we can no longer express the convective state in terms of just three expressions, causing this approximation to become less accurate. However, we have essentially found a small-Rayleigh number approximation for the Nusselt number under free-slip boundary conditions.

This approximate scaling of the Nusselt number holds for small Rayleigh numbers close to the critical Rayleigh number. However, as we increase the Rayleigh number, the Nusselt number begins to diverge from this approximation, regardless of the Prandtl number and domain width. However, we do find that for $5000 \gtrsim Ra \gtrsim 1000$, we find a more accurate approximation in the form

$$Nu = 1 + A\epsilon^x, \text{ where } \epsilon = \frac{Ra - Ra_c}{Ra_c}$$

with different approximate scaling made for each Prandtl number. However, there is a very small difference in these coefficients, meaning that this difference could be due to numerical truncation, as opposed to dependence on the Prandtl number. This is supported by other papers that have found their own approximate scalings [4]. This does not yet scale the maximal Nusselt number against the Rayleigh number. The important point is that the Prandtl number does not have a discernible effect on the scaling approximation.

For the application to Stellar models, we need to investigate higher Rayleigh numbers.

2.2.2 Large Rayleigh numbers

At larger Rayleigh numbers, the domain width begins to play an important role in determining the Nusselt number. Every Rayleigh number has an optimal domain width at which the Nusselt number is maximised. We can then use this to find an approximate scaling between the Rayleigh number and Nusselt number in natural systems. Thus, motivating one of the main problems of this paper: using the adjoint-based maximisation method to optimise the domain width of Rayleigh-Bénard systems to approximate a scaling law between Rayleigh and Nusselt numbers.

It is important to recognise that we are using a simplified model of Rayleigh-Bénard convection by using the Boussinesq approximation and assuming the fluid is incompressible.

However, we can still use this to draw meaningful relationships that apply to the real world. Many papers have attempted to approximate a Nusselt number scaling law for a system using dimensionless constants [4] [11] [3] [2]. However, many are unclear on how they found these maximum Nusselt numbers, further motivating this problem.

As the Rayleigh number increases, the system becomes more turbulent, meaning it needs to be simulated at a higher resolution with smaller time steps to prevent the numerical solver from diverging. Eventually, this reaches a practical limit. Therefore, to predict fluid behaviour at higher Rayleigh numbers, we attempt to extrapolate Nu using an appropriate scaling law.

2.3 Optimising Domain Width

2.3.1 Adjoint-Based Method Application

From here, we can begin to explore how we can apply the adjoint-based method of optimisation to this setup. In this problem, we are interested in the optimal value of L , L^* , that maximises the Nusselt number in the system's steady state. When simulating a fluid, we give it some initial perturbation to create turbulent behaviour. However, after some period of time, the system settles to its stable steady state.

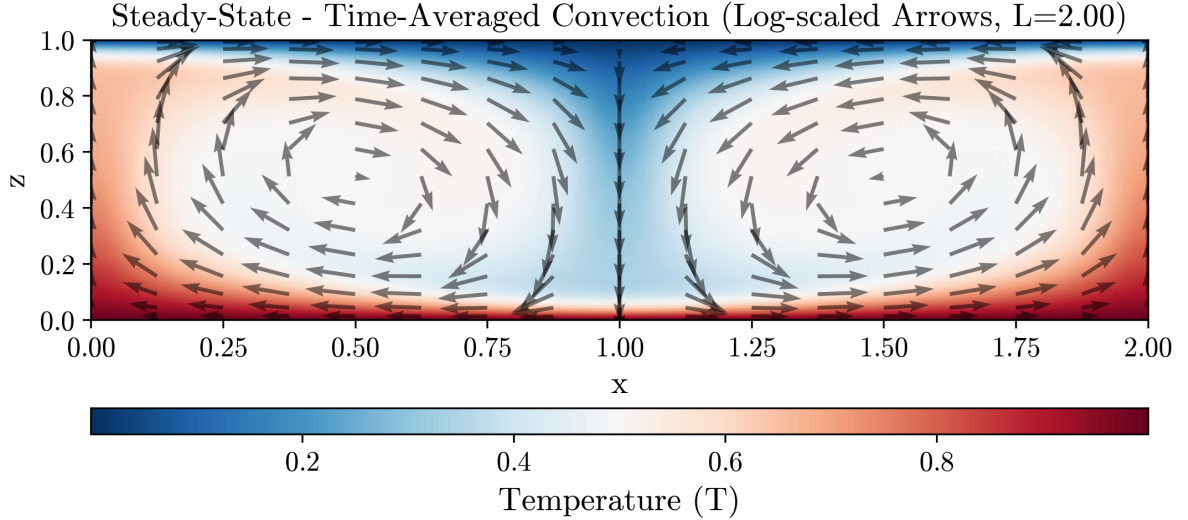


Figure 2.1: Example of a steady-state of a Rayleigh-Bénard system, with $L = 2$ and $Ra = 10^5$ - the arrows are log-scaled velocity vectors.

This domain can fit at most one convection cell (one hot and one cold plume) within its width, creating two rolls between the hot and cold plumes. As the width of the domain changes, these rolls are stretched and squeezed, giving rise to different fluid properties that we will explore

To maximise the Nusselt number and have an objective function in the form we want, we define it as follows:

$$J = 1 + \frac{1}{|\Omega|} \int_{\Omega} v_z T d\Omega$$

where $|\Omega|$ is the unit area/volume of the spatial domain. Through the construction of this problem, we get the final form of the objective function to be:

$$J = \int_0^1 \int_0^L \frac{1}{L} (1 + v_z T) dx dz \quad (2.19)$$

So far, we have defined the system of equations defining our system, as well as the objective function to maximise. Next, we will work out how we can apply the adjoint-based method to optimise the domain width, L , to maximise J . We use free-slip boundary conditions when considering astrophysical flows, setting $v_z(x, 0, t) = v_z(x, H, t) = \mathbf{0}$ and $\frac{\partial v_x}{\partial z}(x, 0, t) = \frac{\partial v_x}{\partial z}(x, H, t) = 0$ (impermeability and zero shear stress). Now we can find the adjoint equations required to solve this problem.

2.3.2 Linearisation

The first hurdle to overcome in constructing this adjoint field is that not all the terms in (2.10) and (2.11) are linear. Therefore, we must linearise each state about its static state, assuming that all terms of quadratic order or higher are negligible when calculating the effect of small perturbations. In practise, this is okay for accuracy as long as we set the discretised time step, δt , to be sufficiently small. Essentially, this offloads some analytical complexity onto the numerical solver. In this problem, our state vector $\mathbf{y} = (\mathbf{v}, T, p)^T$ (and control vector $\mathbf{u} = (L)$), with $\rho = \rho_0$, so we define the linearising transformation:

$$\mathbf{y} = \bar{\mathbf{y}} + \mathbf{y}'$$

This causes our incompressible Boussinesq approximation equations to take a linear form:

$$\nabla \cdot \mathbf{v}' = 0 \quad (2.20)$$

$$\bar{\mathbf{v}} \cdot \nabla \mathbf{v}' + \mathbf{v}' \cdot \nabla \bar{\mathbf{v}} = -\nabla p' + Pr \nabla^2 \mathbf{v}' + Pr Ra T' \hat{\mathbf{e}}_z \quad (2.21)$$

$$\bar{\mathbf{v}} \cdot \nabla T' + \mathbf{v}' \cdot \nabla \bar{T} = \nabla^2 T' \quad (2.22)$$

In our simulations, we will use the time-dependent non-linear equations (2.6)-(2.8), but to analytically find the adjoint field equations, we will use this linear form. Allowing this simulation to run for a long period of time allows the system to eventually settle to a stable steady state. A stable state returns to its equilibrium when perturbed.

This static state satisfies the equations (2.9)-(2.11). We can use this final steady state $(\bar{\mathbf{v}}, \bar{T}, \bar{p})$ to solve for our adjoint field. We now follow a similar procedure as in the examples, and the proof (albeit more complicated) to find the adjoint problem. Let us define the Lagrangian,

$$\begin{aligned} \mathcal{L} = J &+ \int_{\Omega} \boldsymbol{\lambda} \cdot (-\bar{\mathbf{v}} \cdot \nabla \mathbf{v}' - \mathbf{v}' \cdot \nabla \bar{\mathbf{v}} - \nabla p' + Pr \nabla^2 \mathbf{v}' + Pr Ra T' \hat{\mathbf{e}}_z) d\Omega dt \\ &+ \int_{\Omega} \pi (-\nabla \cdot \mathbf{v}') d\Omega dt + \int_{\Omega} \theta (-\bar{\mathbf{v}} \cdot \nabla T' - \mathbf{v}' \cdot \nabla \bar{T} + \nabla^2 T') d\Omega dt \end{aligned}$$

where $(\boldsymbol{\lambda}, \theta, \pi)^T$ is the adjoint field. We use integration by parts and apply the boundary conditions to each term in these integrals to rearrange the terms. We then take advantage of the form $J = \int_{\Omega} F d\Omega$, and then take the first-order variation of $\mathcal{L}$ with respect to $\mathbf{v}'$,

p' and T' respectively to achieve the following adjoint problem.

$$-\bar{\mathbf{v}} \cdot \nabla \boldsymbol{\lambda} + (\nabla \bar{\mathbf{v}})^T \boldsymbol{\lambda} = \nabla \pi + Pr \nabla^2 \boldsymbol{\lambda} - \theta \nabla \bar{T} + \frac{\partial F}{\partial \mathbf{v}'} \quad (2.23)$$

$$\nabla \cdot \boldsymbol{\lambda} = -\frac{\partial F}{\partial p'} \quad (2.24)$$

$$-\bar{\mathbf{v}} \cdot \nabla \theta = \nabla^2 \theta + Pr Ra \lambda_z + \frac{\partial F}{\partial T'} \quad (2.25)$$

where we set the final-time conditions $\lambda(x, T) = 0$ and $\theta(x, T) = 0$. For the spatial boundaries, applying integration by parts with the forward free-slip conditions dictates that the adjoint velocity field must also satisfy impermeability and zero shear stress. Thus, we set $\lambda_z(x, 0, t) = \lambda_z(x, H, t) = 0$ and $\frac{\partial \lambda_x}{\partial z}(x, 0, t) = \frac{\partial \lambda_x}{\partial z}(x, H, t) = 0$. Because the forward temperatures at the boundaries are fixed, the adjoint temperature field remains constrained by Dirichlet boundary conditions, $\theta(x, 0, t) = \theta(x, H, t) = 0$. Once we solve this system, going backwards in time, the adjoint field is used to update the domain width, L .

2.3.3 Re-scaling

Since our objective function depends on our control vector within the bounds of its integral, we cannot directly use the form of the gradient-descent method in the proof. However, the workaround is only a few extra steps. We want to remove this dependence on L in the limits of the integral, meaning we need to scale our x -coordinates, letting $\hat{x} = x/L$. This leads to the transformed objective function,

$$J = \int_0^1 \int_0^1 (1 + v_z T) d\hat{x} dz$$

It also means that we must return to equations (2.10)-(2.11) and (2.20)-(2.22) to transform the x -coordinates throughout the equations. To do this, we define the linear operators

$$\nabla_L = \left(\frac{1}{L} \frac{\partial}{\partial \hat{x}}, \frac{\partial}{\partial z} \right), \quad \nabla_L^2 = \frac{1}{L^2} \frac{\partial^2}{\partial \hat{x}^2} + \frac{\partial^2}{\partial z^2}$$

and apply them to these equations. Thus, we have the final, non-linear, transformed steady-state forward system to be

$$\nabla_L \cdot \mathbf{v} = 0 \quad (2.26)$$

$$\mathbf{v} \cdot \nabla_L \mathbf{v} = -\nabla_L p + Pr \nabla_L^2 \mathbf{v} + Pr Ra T \hat{\mathbf{e}}_z \quad (2.27)$$

$$\mathbf{v} \cdot \nabla_L T = \nabla_L^2 T \quad (2.28)$$

and after linearising each term through a similar method as above, we reach the final transformed, linear adjoint problem

$$-\bar{\mathbf{v}} \cdot \nabla_L \boldsymbol{\lambda} + (\nabla_L \bar{\mathbf{v}})^T \boldsymbol{\lambda} = \nabla_L \pi + Pr \nabla_L^2 \boldsymbol{\lambda} - \theta \nabla_L \bar{T} + \bar{T} \hat{\mathbf{e}}_z \quad (2.29)$$

$$\nabla_L \cdot \boldsymbol{\lambda} = 0 \quad (2.30)$$

$$-\bar{\mathbf{v}} \cdot \nabla_L \theta = \nabla_L^2 \theta + Pr Ra \lambda_z + \bar{v}_z \quad (2.31)$$

subject to the same free-slip adjoint boundary conditions as previously defined in Section 2.3.2. We have successfully transformed our objective function into the appropriate form by shifting the dependence of L into the equations themselves. However, in fixing one problem, we have created another. Due to the way we transformed x , we removed any dependency on L inside the integral, meaning we cannot easily calculate $\frac{dJ}{dL}$. By taking the derivative $\frac{d\mathcal{L}}{dL}$, and applying the chain rule, we find

$$\begin{aligned} \frac{dJ}{dL} = \int_0^1 \int_0^1 \frac{\pi}{L^2} \frac{\partial v_x}{\partial \hat{x}} + \boldsymbol{\lambda} \cdot \left(\frac{v_x}{L^2} \frac{\partial \mathbf{v}}{\partial \hat{x}} + \frac{1}{L^2} \frac{\partial p}{\partial \hat{x}} - \frac{2Pr}{L^3} \frac{\partial^2 \mathbf{v}}{\partial \hat{x}^2} \right) \\ + \theta \left(\frac{v_x}{L^2} \frac{\partial T}{\partial \hat{x}} - \frac{2}{L^3} \frac{\partial^2 T}{\partial \hat{x}^2} \right) d\hat{x}dz \end{aligned}$$

This is not a pleasant integral to solve analytically, so we will solve it numerically, which is more manageable. This is still far more efficient than brute force in this case, and easily scalable to use more control variables.

Therefore, we will numerically solve the forward problem defined in equations (2.26)-(2.28), then solve the backward problem (equations (2.29)-(2.31)). Calculate the surface integral defined above, then iterate through values of L using

$$L_{N+1} = L_N + \alpha \frac{dJ}{dL}$$

The code used to carry out these simulations can be found in my GitHub repository. The code for this problem can be found in the `/AdjointProblems/` folder.

2.3.4 Domain Width Behaviour & Optimisation Results

To find these steady states, we simulate the equations (2.6)-(2.8) with initial conditions $\mathbf{v}(\mathbf{x}, 0) = \mathbf{0}$ and $T(\mathbf{x}, 0) = 1 - z + \delta$, where δ is some small random perturbation to create instability. These are the simplest initial conditions that satisfy the temperature and free-slip boundary conditions. In fact, the $(1 - z)$ -term in this initial condition for temperature is the simplest steady state for our system. However, it does not maximise Nu as it is a purely conductive state ($Nu = 1$); it is also unstable. If we perturb this temperature by some amount, δ , the system becomes unstable and convection begins. We will see later, by optimising the initial conditions, that these initial conditions have no long-term effect on the Nusselt number of the system.

We then run this simulation long enough for the system to settle into a stable state at Prandtl number $Pr = 1$. We take its average state over the latter region of its simulation to estimate the stable state, $\bar{\mathbf{y}}$, and calculate the Nusselt number.

Before we investigate the effectiveness of the adjoint method, we can plot the Nusselt number against the domain width to build some intuition on the behaviour of a fluid as we increase the domain width.

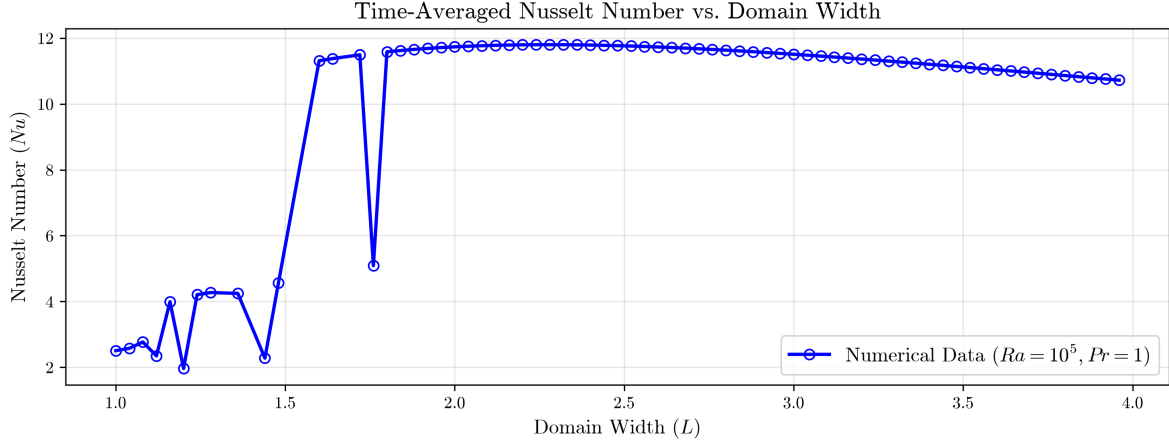


Figure 2.2: Plot of Nusselt number against domain width for $Ra = 10^5$ with free-slip boundary conditions and $Pr = 1$.

We have a single global maximum in the optimal domain width for this Rayleigh number; we aim to find the Nusselt number at this optimal domain width. However, this will require care as there is a drop-off in the Nusselt number into a region of many local maxima. In this region, the domain width is too small for the fundamental mode, with wave number k_1 , to fit inside, causing the amount of heat transport to drop. Our spectral solver causes the chaotic behaviour as heat transport becomes dominated by higher-order wave numbers.

In an unconfined system, the Nusselt number acts as a continuous function of the horizontal wave number, k_n , reaching a global maximum at an optimal, physically preferred wave number, $k_{opt} = 2\pi/L^*$. Approximating this local maximum with a second-order Taylor expansion yields a quadratic efficiency penalty for any deviation from this optimal state, such that $\Delta Nu \propto (k - k_{opt})^2$. However, due to the finite domain width L , the numerical solver restricts the fluid to discrete Fourier modes spaced strictly by $\Delta k = 2\pi/L$. The maximum possible efficiency drop occurs when k_{opt} lands exactly halfway between two allowable modes, forcing a maximum displacement of $|k - k_{opt}| = \pi/L$ (at $L = 1.76$). Substituting this maximum displacement into the quadratic penalty reveals that the worst-case drop in the Nusselt number decays at a rate proportional to $1/L^2$. As the domain width increases, the allowable Fourier modes pack densely together in wave number space. This increasing density of states guarantees that an available discrete mode will always fall sufficiently close to k_{opt} , causing the artificial geometric penalty to rapidly vanish and allowing the fluid to transition seamlessly between multi-cell states.

Our spectral solver causes this behaviour, meaning that we need to take additional care when applying our adjoint-based maximisation method. That is, have an initial guess, L_0 , sufficiently large as to avoid dropping into this chaotic region; otherwise the gradient ascent method will not work. To mitigate the extra time taken to complete the optimisation, we can start with an aggressive learning rate, α , then if the next Nusselt number is smaller than the previous (meaning we've dropped into this chaotic region), we halve α .

Considering these factors, we can run the adjoint-based maximisation for a given Rayleigh number.

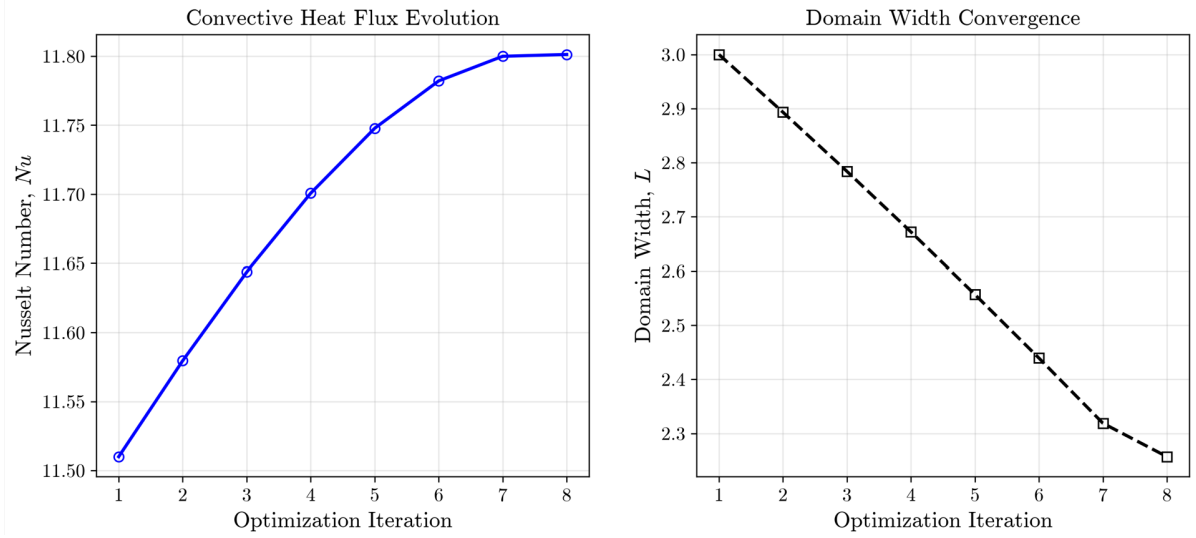


Figure 2.3: Optimisation results under free-slip boundary conditions for $Ra = 10^5$, $Pr = 1$ and $L_0 = 3$.

These results align with our intuition from Figure 2.2, where we converge to this maximal $Nu = 11.8011$ at $L = 2.255077$. Thus, the adjoint-based maximisation of the Nusselt number has been successful in converging. To further understand our spectral solver and the mathematical meaning of this optimal solution, we can analyse the individual modes that make up these optimal steady state solutions.

2.3.5 Mode Analysis

Our numerical solution in the x -direction comprises many Fourier modes. Each with its own associated wavenumber $k_n = \frac{2\pi n}{L}$ and coefficients, meaning that each mode is responsible for a proportion of the convective heat transfer throughout the system. We can investigate which modes drive most convective heat transport as the domain width changes.

Focusing on $Ra = 10^5$, we will compare turbulent, stable, and optimal convection to understand how the roles of these modes change. We will also investigate which modes transport the most heat at larger values of L - such as integer multiples of the optimal domain width - where more than one convective cell forms. To analyse each mode, we apply a Real Fast Fourier Transform (RFFT) to the solution to decompose it into a series of combinations sine and cosine waves, each combination having an associated wave number, $k_n = \frac{2\pi n}{L}$; and taking its average convective heat flux $\langle wT \rangle$ across the domain. This creates a plot known as a Heat Flux Spectrum.

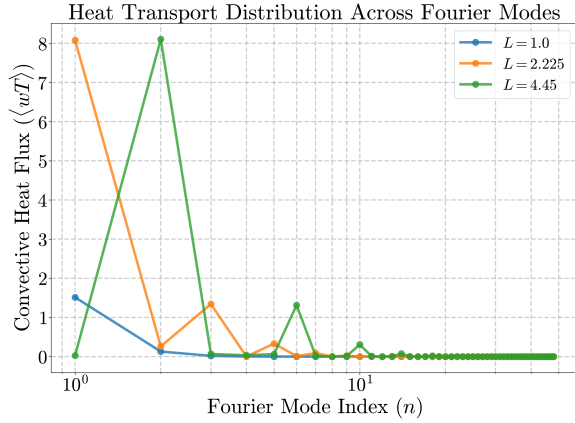


Figure 2.4: Heat Flux Spectrum for values of L with $Ra = 10^5$.

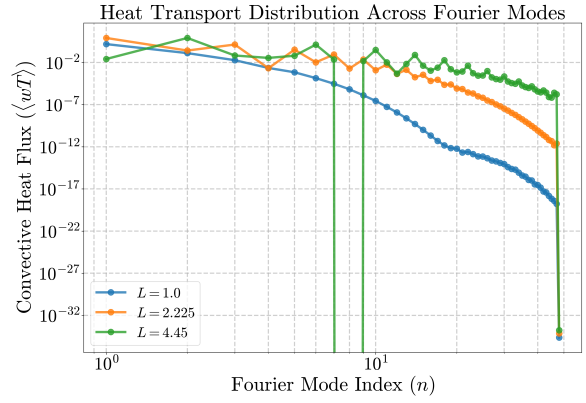


Figure 2.5: Heat Flux Spectrum (log-scaled) for values of L with $Ra = 10^5$.

At integer, n_0 , multiples of the optimal domain width (at which we achieve the same maximal Nusselt number), the dominant mode has the wave number k_{n_0} . The only active modes have wave numbers $n_0(2n - 1)$, making it easy to simplify models at these lower Rayleigh numbers. This can also help create more advanced spectral solver configurations by creating smarter dealiasing filters and sparse matrix structures - effectively ignoring the predictable space between these spikes. By simplifying the model to only include these modes of smallest index, we could explain the majority of the behaviour of our system. In particular, at these multiples of L^* , we would achieve the most accurate simplified models.

Having shown that we can optimise the domain width at a given Rayleigh number, we can now apply this to a list of Rayleigh numbers to build an approximate scaling between the Nusselt number and Rayleigh number.

2.3.6 Scaling the Nusselt number for higher Rayleigh numbers

For a given Rayleigh number, we can now find the optimal domain width and the maximal Nusselt number, which can be used to find an approximate scaling between the Nusselt number and Rayleigh number for higher Rayleigh numbers.

Sondak et al. have their own approximate scaling law between Nu and Ra for no-slip boundary conditions, following $Nu = 0.115Ra^{0.31}$ for $Ra > 10^5$. This was calculated from data when $Pr = 7$, a typical property of seawater. However, they also found no discernible difference in scaling laws from each Prandtl number - similar to our findings in Section 2.2.1. Therefore, by carrying out the adjoint-based maximisation to maximise the Nusselt number, we can find our own scaling for free-slip boundary conditions.

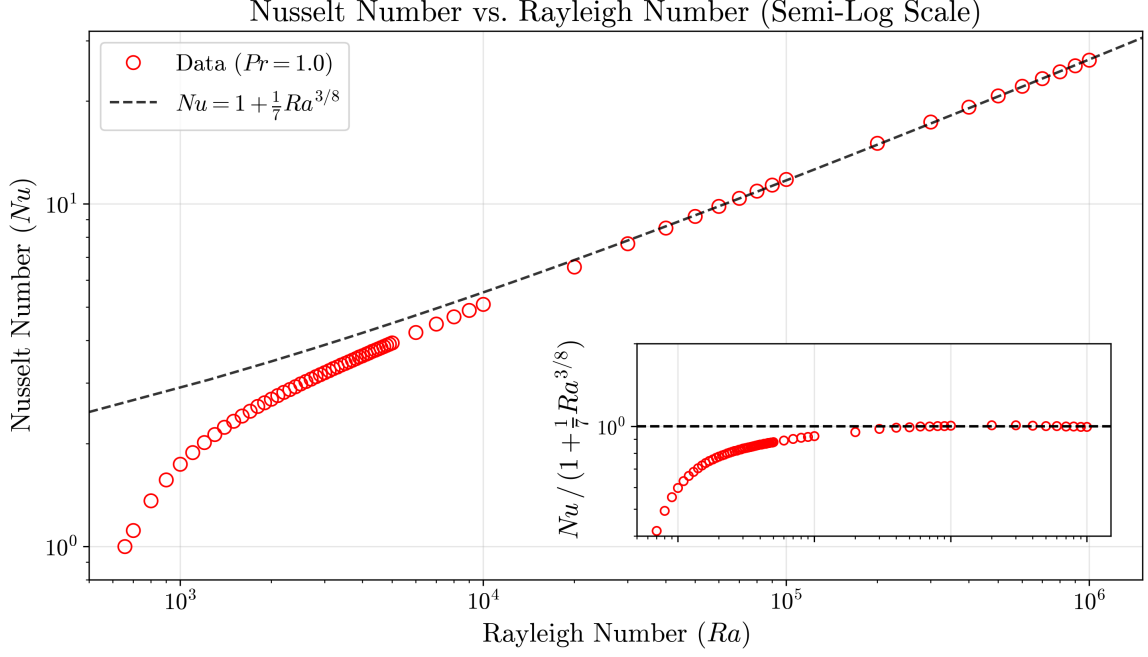


Figure 2.6: Optimal Nu against Rayleigh number under free-slip boundary conditions, with $Pr = 1$.

We find an approximate asymptotic scaling of

$$Nu \sim 1 + \frac{1}{7}Ra^{3/8} \quad (2.32)$$

This is the main result this paper aimed to find; it can be used to extrapolate properties of stellar media with Rayleigh numbers too high for practical simulations, assuming this scaling is independent of the Prandtl number.

From here, one may wish to investigate other properties of astrophysical fluids that this links to, such as viscous dissipation.

2.3.7 Viscous Dissipation

A topic of interest for many researchers in fluid dynamics is the concept of viscous dissipation within a fluid. It is a measure of how much the velocity of a fluid changes across distance, $\frac{dv_i}{dx_i}$; that is, how much energy is lost to internal friction. This is directly linked to the kinetic energy of the system by the conservation of energy. Maximising viscous dissipation also maximises the kinetic energy within the system. We are interested in how this viscous dissipation changes with domain width, and whether maximising the Nusselt number has a relationship with this viscous dissipation.

The viscous dissipation is defined as

$$\begin{aligned} \epsilon(\mathbf{x}, t) &= \nu(\partial_i v_j(\mathbf{x}, t))^2 \\ \epsilon(\mathbf{x}, t) &= \nu \left[\left(\frac{\partial v_x}{\partial x} \right)^2 + \left(\frac{\partial v_x}{\partial z} \right)^2 + \left(\frac{\partial v_z}{\partial x} \right)^2 + \left(\frac{\partial v_z}{\partial z} \right)^2 \right] \end{aligned}$$

To investigate its behaviour, we will take its average over the steady-state solution of our system for each value of L we iterate over. Therefore, we will monitor the quantity

$$\langle \epsilon \rangle_{\Omega} = \frac{1}{|\Omega|} \int_{\Omega} \epsilon(\mathbf{x}, t) d\Omega$$

The relationship between L and ϵ has been studied in other papers [3] - for no-slip boundary conditions, the hypothesised behaviour is as follows.

$$\epsilon = \frac{\nu^3}{L^4} (Nu - 1) Ra Pr^{-2}$$

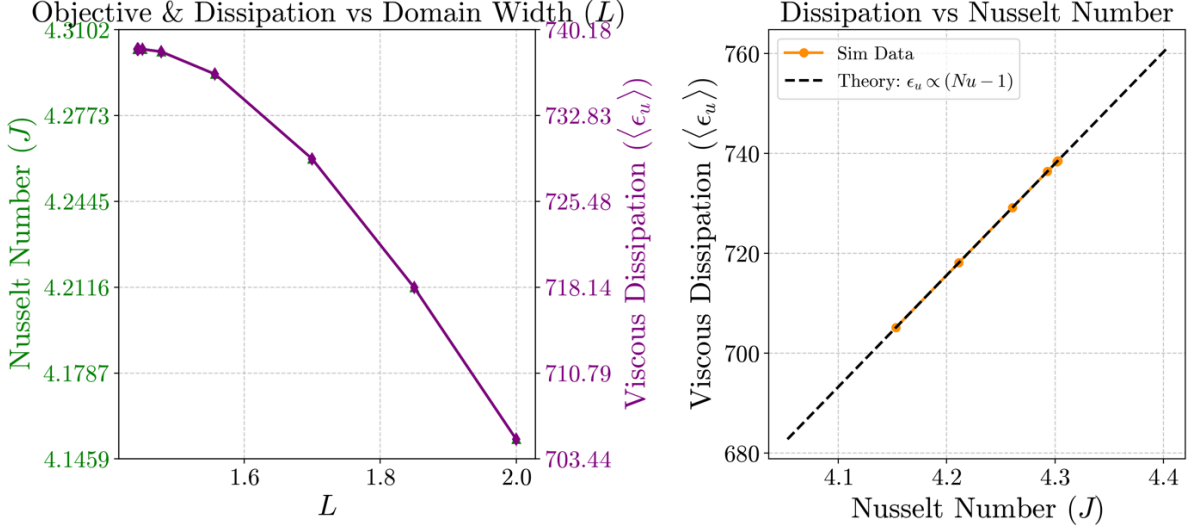


Figure 2.7: Plots of the Nusselt number and viscous dissipation against domain width, L , for $Ra = 5 \times 10^4$, $Pr = 1$ under no-slip boundary conditions.

The viscous dissipation does not follow this same analytical behaviour under free-slip boundary conditions. We still observe this proportionality between the Nusselt number and viscous dissipation. However, under free-slip boundary conditions, we find that viscous dissipation now follows

$$\epsilon = \sqrt{RaPr} (Nu - 1) \quad (2.33)$$

Thus, the optimal domain width minimises the amount of turbulence in the system. This turbulence is caused by the Fourier modes with smaller wavelengths (larger wave numbers), aligning with our previous results in Section 2.3.5. Maximising the Nusselt number with respect to the domain width also maximises the viscous dissipation, and thus, the kinetic energy of our system.

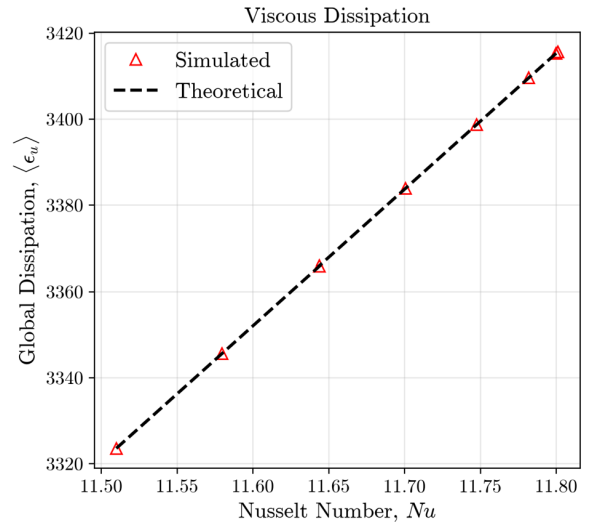


Figure 2.8: Viscous dissipation against Nusselt number under free-slip boundary conditions, following results in Figure 2.3.

2.4 Optimising Initial Conditions

We have optimised a single parameter, the domain width, to maximise the Nusselt number, Nu . Returning to a case similar to our proof and the two-dimensional example, we are now interested in how the initial conditions affect the Nusselt number. In the previous problem, we said they had no long-term impact on the Nusselt number, but we will now prove this.

This is a $4N_xN_z$ -parameter problem, as each point in our discretised domain has its own values for velocity in the x - and z -directions, temperature and pressure. This problem aims to highlight the major advantages of the adjoint-based method.

2.4.1 Problem Setup

We will use the same problem setup as in equations (2.6)-(2.8) for the forward problem, and equations (2.23)-(2.25) for the adjoint problem. The main change we have is $\nabla_{\mathbf{u}}J$ where $\mathbf{u}$ is now our initial conditions, the same as our definition Section 1.1.

In our proof, we found a general form to calculate $\nabla_{\mathbf{u}}J$, with our state vector $\mathbf{y} = (\mathbf{v}, T, p)^T$. Seen as $\mathbf{u}(\mathbf{x})$ is exactly our initial conditions, we have $\mathcal{B}$ to be the identity matrix, meaning $\mathcal{B}^*$ is also the identity. Due to our objective function including v_z , we face a similar issue as in our two-dimensional example where the optimiser will make v_z very large to make Nu very large. Hence, we impose conservation of kinetic energy within our system, so after each iteration we rescale the velocity field to conserve the kinetic energy between iterations. This requires an initial guess making $\mathbf{v}(\mathbf{x}, 0) = \mathbf{0}$ and $T(\mathbf{x}, 0) = H - z + \delta$. We then calculate the kinetic energy of the system at the end of this first forward pass and take that as the constant kinetic energy of the system for subsequent iterations. This is not the only way to restrict the magnitude of the initial conditions, though. We then use the appropriate gradient ascent algorithm to update our initial conditions on each iteration.

$$\mathbf{u}_{n+1} = \mathbf{u}_n + \alpha \begin{pmatrix} \lambda \\ \theta \\ \pi \end{pmatrix} (\mathbf{x}, 0)$$

where the adjoint problem is run from $t = T$ to $t = 0$. Given our previous results in Sections 2.3.1 and 2.3.4, intuition might suggest that the optimal initial conditions for maximising the (time-averaged) Nusselt number would be the steady-state that the system settles to over time. This steady state minimises the fluctuations in the Nusselt number at the beginning of the simulation, so one might expect the optimal initial conditions to also minimise these fluctuations by becoming this steady state.

2.4.2 Initial Conditions Optimisation Results

Using the Dedalus framework, we can solve this adjoint problem, maximising the Nusselt number, achieving the following initial conditions throughout the iterative process.

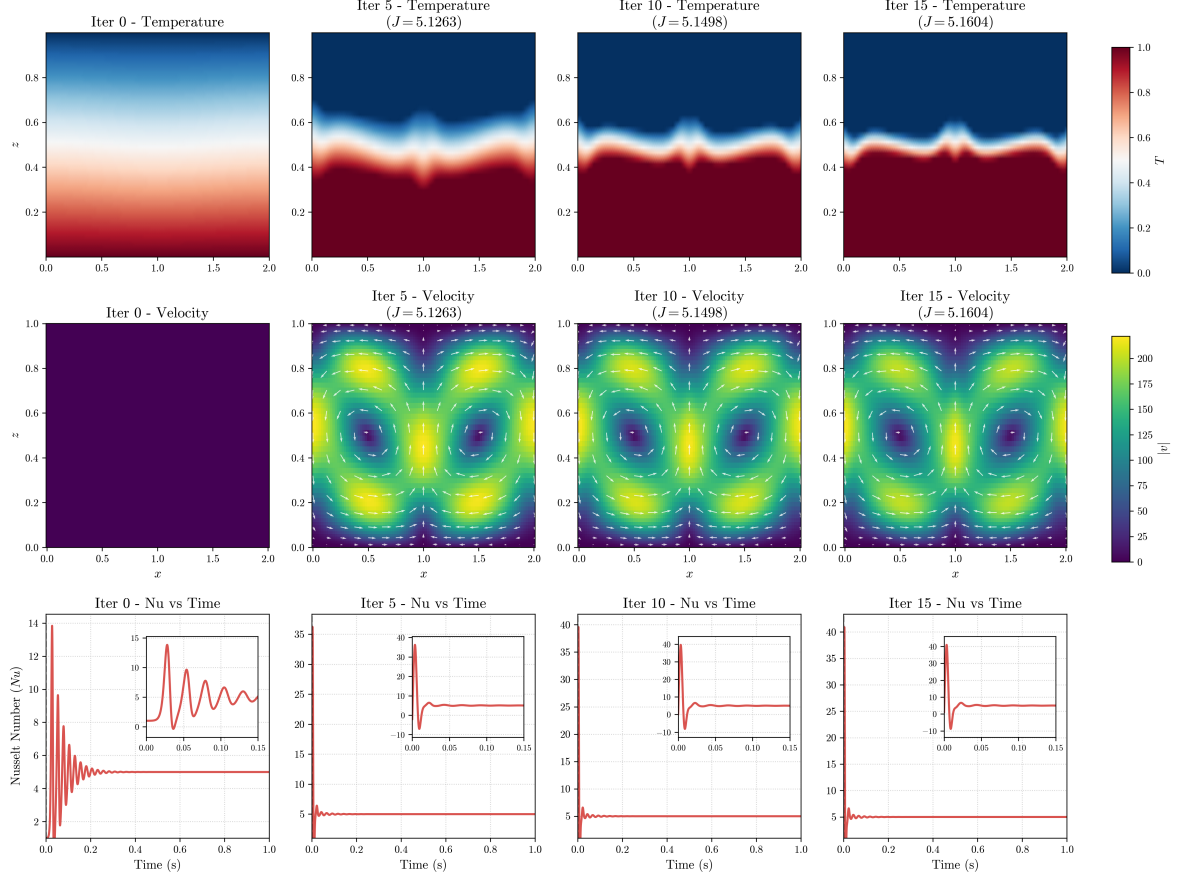


Figure 2.9: Evolution of initial conditions optimisation to maximise the (time-averaged) Nusselt number of a Rayleigh-Bénard system with $L = 2$, $Ra = 10^5$, $Pr = 1$, and no-slip boundary conditions.

The velocity field is in the form we would intuitively expect, matching the steady-state solution with these two counter-rotating convection rolls. However, the temperature field is not in the same form as the steady-state solution it settles to - creates this steep temperature gradient across the middle of the domain. Notice that this configuration reduces the fluctuations in the Nusselt number at the beginning of the simulation. That is, these initial conditions that maximise the time-averaged Nusselt number across the simulated time also minimise the time taken for the system to settle to a stable steady state.

For both no-slip and free-slip boundary conditions, we find that regardless of the initial conditions, the system always settles to the same steady state. The initial conditions affect only the short transient phase at the beginning of the simulation. However, the long-term (as $t \rightarrow \infty$) state of the system is unaffected by these initial conditions. This optimisation method simply maximises the time-averaged Nusselt number in this beginning transient phase, and minimises the time taken to settle to this stable steady state.

In light of this, our previous analysis for maximising the Nusselt number with respect to the domain width merely maximises the Nusselt number that the system converges to, regardless of the initial conditions - so long as we allow the simulation to run for long enough to allow the system to converge.

Chapter 3

Adding Rotation

In this chapter we focus on the impact of the Coriolis forces on the critical Rayleigh number under free-slip conditions. We will not apply the adjoint-based maximisation due to time constraints. However, this chapter provides the appropriate problem setup to then apply the adjoint-based maximisation method to an appropriate problem.

Using a similar setup outlined in G. Hadjerci et al. 2026 [5], we can introduce the effects of rotation to our fluid, typically found in stellar interiors and planetary oceans and atmospheres. Rotation typically inhibits convection, causing the critical Rayleigh number, Ra_c , to increase with the rate of rotation [12], ω .

There are many studies in various fields studying the Coriolis force and its impact in nature, but we will simply introduce the non-dimensional constant, the Taylor number, Ta . This is a measure of how much we are rotationally driving the system.

$$Ta = \frac{4\omega^2 H^4}{\nu^2}$$

The Coriolis force is defined by

$$\mathbf{F}_c = -2\omega(\hat{\mathbf{l}} \times \mathbf{v})$$

where $\hat{\mathbf{l}}$ is the unit vector in the direction of the axis of rotation, and ω is the angular velocity. In our case, we will take this to be $\hat{\mathbf{e}}_z$. It also needs to be non-dimensionalised through scaling by the thermal diffusion time, H^2/κ . This gives rise to Newton's Second Law of our system, including rotation.

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} + Pr\sqrt{Ta}(\hat{\mathbf{e}}_z \times \mathbf{v}) = -\nabla p + Pr\nabla^2 \mathbf{v} + PrRaT\hat{\mathbf{e}}_z \quad (3.1)$$

We can now apply similar Numerical methods to observe how rotation affects the formation of convective cells in our Rayleigh-Bénard problem. Upon adding this force, we do need to consider the y -component of our velocity in the numerical solver.

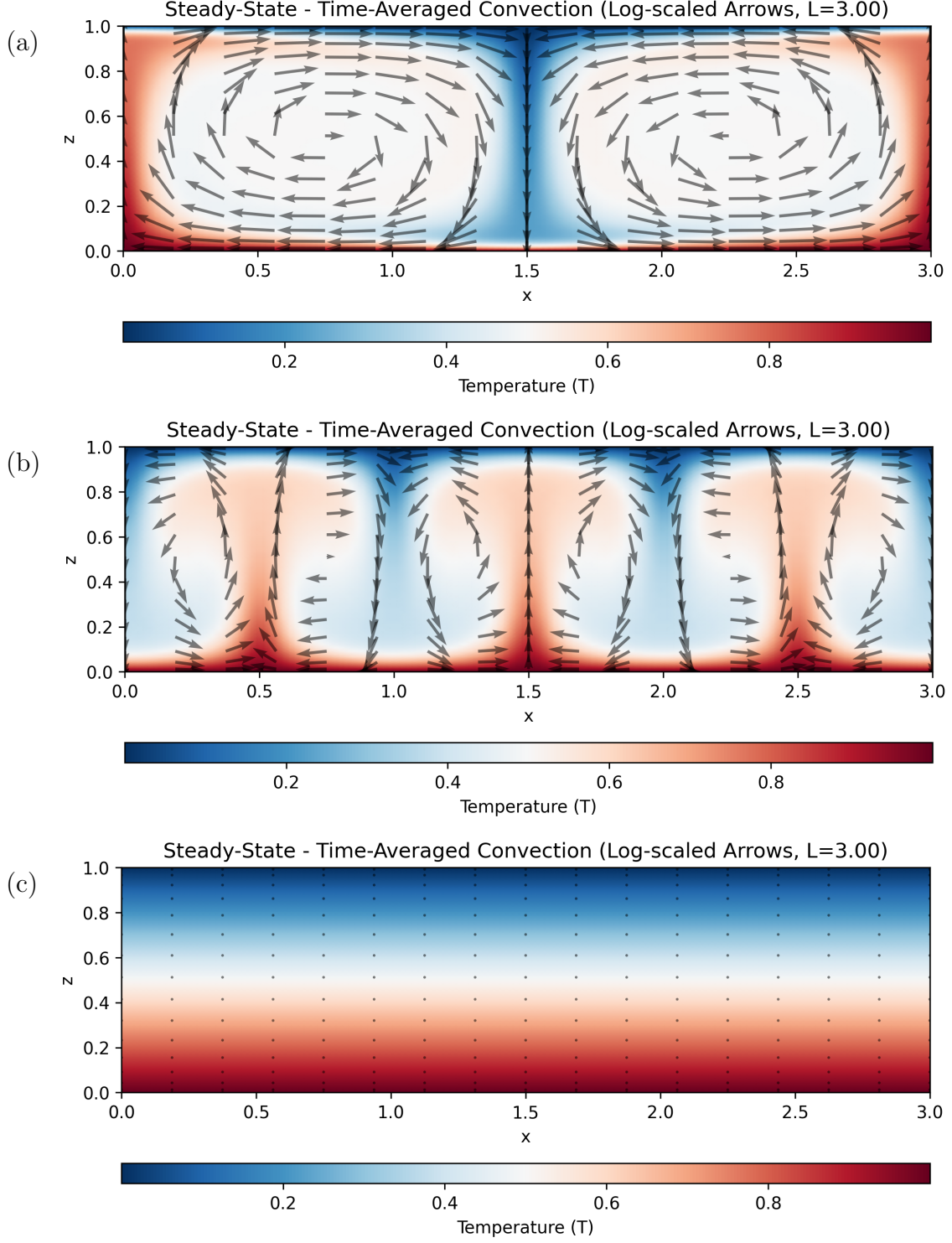


Figure 3.1: Time-averaged solution for a rotating system of Rayleigh-Benard convection with $Ra = 10^5$, $Pr = 1$, $L = 3$ with varying rotation (a) $Ta = 0$, (b) $Ta = 10^5$, (c) $Ta = 10^7$ and free-slip boundary conditions. The arrows represent the (log-scaled) velocity vector field.

In fact, we can analytically prove how Ra_c scales with Ta using a similar method as in Section 2.2.1. By linearising Equation (3.1) about the purely conductive state of the

system, when $Ra = Ra_c$, we have $\bar{T} = 1 - z$ and $\bar{\mathbf{v}} = \mathbf{0}$. Hence,

$$Pr\sqrt{Ta}(\hat{\mathbf{e}}_z \times \mathbf{v}') = -\nabla p + Pr\nabla^2 \mathbf{v}' + PrRa_c T \hat{\mathbf{e}}_z \quad (3.2)$$

$$-v'_z = \nabla^2 T' \quad (3.3)$$

Similarly as before, we take the curl of (3.2) to find the vorticity equations

$$\nabla^2 \zeta = -\sqrt{Ta} \frac{\partial v'_z}{\partial z} \quad (3.4)$$

$$\nabla^4 v'_z + Ra_c \nabla_H^2 T' - \sqrt{Ta} \frac{\partial \zeta}{\partial z} = 0 \quad (3.5)$$

where $\nabla_H^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$. If we assume $v'_z \propto \sin(\pi z)e^{ikx}$, we can replace $\nabla_H^2 \rightarrow -k^2$ and $\nabla^2 \rightarrow -(k^2 + \pi^2)$. Imposing $v'_z = 0$ and $\frac{\partial v'_z}{\partial z} = 0$ at the boundaries and by substituting into (3.3) and (3.4) we get

$$T' = \frac{v'_z}{k^2 + \pi^2} \quad \text{and} \quad \zeta = \frac{\sqrt{Ta} \frac{\partial v'_z}{\partial z}}{k^2 + \pi^2}$$

Substituting these back into (3.5) we have

$$(k^2 + \pi^2)^2 v'_z - Ra_c \frac{k^2}{k^2 + \pi^2} v'_z + Ta \frac{\pi^2}{k^2 + \pi^2} v'_z = 0 \quad (3.6)$$

$$\implies Ra_c = \frac{(k^2 + \pi^2)^3 + \pi^2 Ta}{k^2} \quad (3.7)$$

Notice that in the unrotating case, $Ta = 0$, this form is the same as the form we found in Section 2.2.1 (Equation (2.17)). Minimising with respect to the wave number, we have

$$\begin{aligned} \left. \frac{dRa_c}{dk} \right|_{k=k_c} &= \frac{(4k_c^3 - 2k_c\pi^2)(k_c^2 + \pi^2)^2 - 2k_c\pi^2 Ta}{k_c^4} \\ \implies \pi^2 Ta &= (k_c^2 + \pi^2)^2 (2k_c^2 - \pi^2) \end{aligned}$$

For a given Taylor number, we can solve this cubic for the critical wave number and solve for the critical Rayleigh number. We can do this using Cardano's formula to achieve

$$\begin{aligned} k_c &= \pi \sqrt{\frac{1}{2} \left[\sqrt[3]{1 + \frac{2Ta}{\pi^4} + 2\sqrt{\frac{Ta}{\pi^4} \left(1 + \frac{Ta}{\pi^4}\right)}} + \sqrt[3]{1 + \frac{2Ta}{\pi^4} - 2\sqrt{\frac{Ta}{\pi^4} \left(1 + \frac{Ta}{\pi^4}\right)}} - 1 \right]} \\ k_c &= \pi \sqrt{\frac{1}{2} [\Psi(Ta) - 1]} \implies k_c^2 = \frac{\pi^2}{2} [\Psi(Ta) - 1] \end{aligned}$$

which we can substitute into (3.7) to find a complete analytic expression for the critical Rayleigh number in terms of the Taylor number. Again, notice that $Ta = 0$ still gives the same k_c found in the non-rotating case. Thus, we find the expression

$$Ra_c = \frac{\frac{\pi^4}{4} [\Psi + 1]^3 + 2Ta}{\Psi - 1} \quad (3.8)$$

Many papers [5] have investigated the behaviour of a rapidly rotating system, as $Ta \rightarrow \infty$. This could mean taking $\omega \rightarrow \infty$ or $\nu \rightarrow 0$. We find this by taking the first-order expansion of Ψ to find that $\Psi(Ta) \approx (4\frac{Ta}{\pi^4})^{1/3}$, giving the final approximate asymptotic scaling

$$Ra_c \approx 3 \left(\frac{\pi^4}{4} \right)^{1/3} Ta^{2/3} \quad (3.9)$$

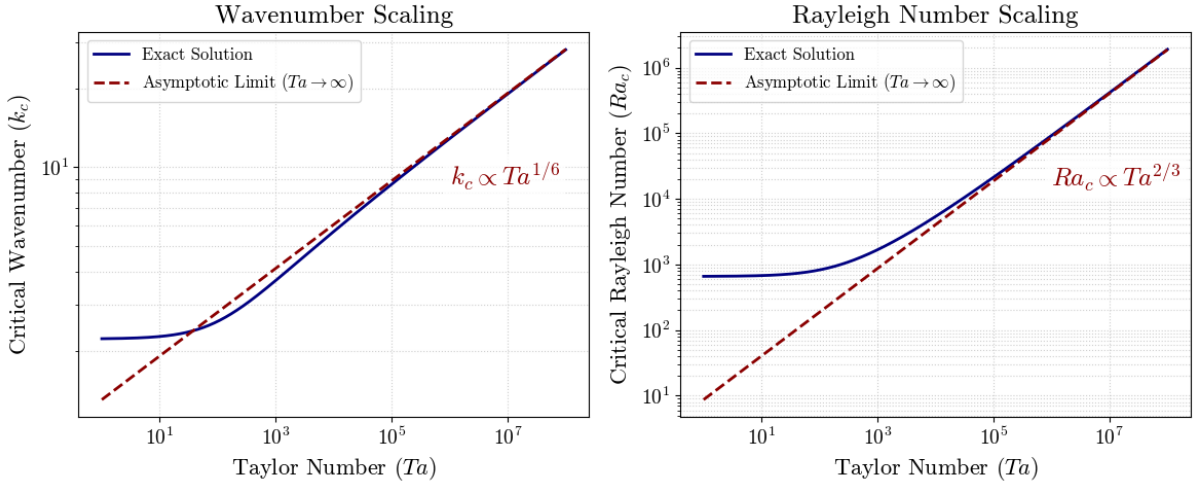


Figure 3.2: Plots of the critical wave number, k_c , and Taylor number against the critical Rayleigh number, Ra_c , with asymptotic behaviour as $Ta \rightarrow \infty$ under free-slip boundary conditions.

Thereby, showing how rotation inhibits convection, increasing the critical Rayleigh number required for convection to occur. Thereby decreasing the Nusselt number of the system; further analysis of this would be needed to draw more formal conclusions.

Although we have not applied the adjoint-based maximisation method to a rotating case, the methods explored in this paper could be applied to the rotating case.

At this point, we have reached the end of this research project. However, natural extensions from this project involve the application of the adjoint-based maximisation method to problems involving rotation, magnetic fields and non-constant density. This could lead to higher-dimensional, more accurate scaling laws to better predict the behaviour of fluids within stellar bodies. One could also model these systems in a polar or spherical basis to model astrophysical fluids on a large scale. These complexities require extensive computational resources that we do not (currently) have access to. However, we can still make valuable observations from the simpler models used in this paper.

Chapter 4

Conclusions

In this paper, we have successfully demonstrated the power of adjoint-based optimisation to rigorously maximise convective heat transfer within Rayleigh-Bénard systems. By leveraging the continuous adjoint equations alongside the Dedalus spectral solver, we transitioned beyond traditional brute-force parameter sweeps, providing a highly efficient computational framework for discovering optimal fluid states.

A primary achievement of this research was the optimisation of the spatial domain width to strictly maximise the Nusselt number, Nu . By restricting the fluid to its most efficient geometric state, we eliminated the chaotic competition of higher-order Fourier modes, which in turn minimised internal turbulence and maximised global viscous dissipation. This mathematically coerced optimal state allowed us to isolate and extract a clean, theoretical asymptotic scaling law for free-slip boundaries, $Nu \sim 1 + \frac{1}{7}Ra^{3/8}$. This fractional relationship provides a powerful predictive tool for extrapolating the behaviour of stellar media in high-Rayleigh regimes that currently sit beyond practical simulation limits.

Furthermore, the application of the adjoint method to the system's initial conditions yielded crucial insights into the transient dynamics of astrophysical fluids. We mathematically verified that while optimal initial perturbations effectively minimise the time required for the system to settle, they have no permanent bearing on the final, long-term steady state of the convection rolls.

Finally, by introducing the Taylor number, Ta , we established the analytical foundation for rotating convective systems, proving exactly how the Coriolis force fundamentally inhibits heat transfer by elevating the critical Rayleigh number required for the onset of convection. While the finite memory and processing constraints of high-performance computing clusters currently limit the expansion of these adjoint solvers into fully three-dimensional, rapidly rotating environments, the methodologies validated in this study establish a highly robust foundation. Natural extensions of this work into polar or spherical coordinate bases, alongside the introduction of magnetic fields and compressible density profiles, will further bridge the gap between idealised mathematical models and the immense, turbulent realities of stellar interiors.

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